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A machine learning approach combining expert knowledge with genetic algorithms in feature selection for credit risk assessment

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ABSTRACT

Most credit scoring algorithms are designed with the assumption to be executed in an environment characterized by an automatic processing of credit applications, without considering the input of expert opinions. Since in credit scoring applications expert opinions have been proved very helpful, in this work, we propose a combination strategy of integrating soft computing methods with expert knowledge. The ability of interpretation of the predictive power of each feature in the credit dataset is strengthened by the engagement of experts in the credit scoring process, and the proposed wrapper-based feature selection approach which explores how the features contributing most towards the classification of borrowers. In particular, an unsupervised machine learning algorithm allows experts to create one or more clustering scenarios regarding the features by defining a desired number of clusters per scenario. For each clustering scenario, the Analytic Hierarchy Process is applied for helping experts to make preferences for features of each cluster, set pair-wise constraints for features of equal importance, and evaluate their subjective judgments in terms of consistency. Then, expert opinions are taken into consideration for solving a credit scoring problem in the form of an optimization problem subject to constraints via soft computing methods based on supervised machine learning and evolutionary optimization algorithms. Testing instances on standard credit dataset are established to verify the effectiveness of the proposed methodology.

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1. Introduction

In recent years, there has been an increase in studies on credit risk assessment in the banking industry. Credit default risk is a special type of credit risk that arises when a borrower is not able to make his/her required credit (e.g., loan) payments [1]. Usually, credit default risk is evaluated by banks before approving any credit (e.g., credit card, consumer loan, mortgage loan, etc.) via a credit default risk assessment/evaluation process where the emphasis is placed on the five C's of the borrower: (i) the credit history, (ii) the capacity to repay (i.e., the debt-to-income ratio), (iii) the capital, (iv) the loan's condition (i.e., the purpose of the loan, the loan amount, interest rates, etc.) and (v) the associated collateral [2].

One of the most popular credit risk assessment tools is credit scoring [1–3]. Credit scoring is an important topic in the financial risk management area and it has been the major focus of the banking industry, especially in the aftermath of the financial crisis of 2007-2008 [4]. Several statistical and machine learning (ML) models have been introduced in the literature to evaluate the credit-worthiness of borrowers [5]. The performance of these models has been investigated by applying them to datasets that reflect traditional banking or pee-to-peer (P2P) lending environments. P2P lending can be described as a new, online based financial intermediary that connects people willing to borrow with people willing to lend their money [6]. On the other hand, credit scoring modeling in traditional banking environments aims at solving corporate credit scoring problems (e.g., [7, 8]) and credit scoring problems with (e.g., [9]) and without collaterals (e.g., [10]). It is worth mentioning that there are three main ways of modeling credit scoring problems: (a) the traditional credit scoring that is based on data from previously granted credit applications, where the borrowers' repayment behavior has been analyzed and explored (e.g., [11], (b) the reject-inference based credit scoring which aims at estimating the risk of defaulting for loan applications that have been rejected under a specific banking acceptance policy (i.e., improve the quality of a scorecard using data contained in rejected loan applications) (e.g., [12]), as well as (c) the profit-based credit scoring which aims at maximizing the profit of a credit scoring model by balancing the benefits and losses of granting credit with the variable acquisition costs (e.g., [13,14]).

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In this direction, several computational algorithms for developing efficient scoring models have been proposed, whereas a special emphasis have been put on the management of imbalanced datasets (e.g., [15,16]), the issues of variable selection and the selection of training and test datasets (e.g., [17]), as well as the explainability of the machine learning algorithms (i.e., the ability to explain machine learning model's autonomous decisions and actions to human users) (e.g., [18]). In addition to this, some interesting critical analyses have been published which review the role of some default tasks in nowadays credit scoring modeling such as the process of segmentation [19] and the usage of demographic data [20]. The main outcomes of the aforementioned analyses indicate that grouping customers into several groups and building scorecards for them do not always improve the model performance in credit scoring, while scoring models consisting of e-mail usage, psychometric and demographic variables can give greater predictive accuracy than models which are based only on demographic data. It should be noticed that, since these models are data-driven, there is no a unique method that can optimally solve any credit scoring problem. Mining multi-dimensional databases to capture reliable, valuable and accurate information, as well as addressing big data challenges and difficulties (e.g., data imperfection, data inconsistency, data confliction, data heterogeneity, etc.) due to its "5Vs" (i.e., volume, velocity, variety, veracity and value) are two of the most significant research topics nowadays [21,22]. For these general reasons, several methodological approaches using the concept of learn-heuristics [23] have been introduced in the literature, combining meta-heuristic optimization algorithms and machine learning methods to solve data-driven and feature-selection based problems (e.g., [24–28]).

According to Dastile et al. [5], despite the advances and applications of ML models in credit scoring literature, there still exist some limitations. Issues such as the usage of macro-economic variables, the application of an exploratory data analysis approach to handle missing values and outliers, the design of feature selection and feature extraction mechanisms, the usage of different cut-offs for classifying borrowers, the potential correlation between variables, and the incorporation of Type II error are some practical examples. Moreover, Costello et al. [29] mentioned the significance of modeling human's engagement in the lending process, whereas Pławiak et al. [30] noticed that four main issues should be considered when the credit-worthiness of borrowers is evaluated: (a) feature extraction and feature selection mechanisms, (b) the classification algorithm selection process, (c) the optimization of models' parameters, and (d) the evaluation of the outcome.

Most of the research papers related to credit scoring in the banking industry propose computational intelligence approaches (e.g., machine learning and optimization techniques) that may be executed efficiently and fast in an environment characterized by an automatic processing of credit applications. However, the credit approval systems of banks are not always fully automated. Usually, for credit such as a mortgage loan or a consumer loan banks require the contribution from a financial analyst (or credit officer) who combine his/her knowledge and system's proposals (e.g., computed indicators, demographic and financial information) to approve or reject a credit application. Combining expert knowledge with computational intelligence approaches in the credit risk assessment process has not been extensively researched in the literature (e.g., [31,32]). In addition to this, several learning models such as neural networks face several problems due to their "black-box" nature as they cannot successfully explain their computational results in a language close to a human expert (i.e., they transform decision variables using weights and activation functions into hidden value space) [33,34].

Considering the aforementioned issues, the main goals of this study can be summarized as follows:

- To assist banks in the task of credit risk evaluation when they use semi-automated loan approval systems.
- To provide experts with computational methods that are able to combine computational intelligence and human-understandable linguistic inference model (i.e., expert opinions) in credit scoring applications.
- To define a feature selection process subject to constraints that can be determined by experts with respect to their experience using multi-criteria decision analysis tools and evolutionary optimization algorithms.
- To verify the effectiveness of the proposed methodology by applying it to a well-known credit risk dataset.

As to the main contribution and novelty of this work, we introduce a new two-phase methodology combining expert knowledge with evolutionary feature selection in an optimization framework for credit risk assessment. Our goal is not to improve the performance of classical machine learning algorithms as it happens in many studies, but to facilitate the engagement of experts in the solution of credit scoring problems. In the context of the framework, unsupervised machine learning algorithms (e.g., the K-Means algorithm for clustering purposes), supervised machine learning algorithms (e.g., the k-Nearest neighbors — KNN- algorithm and the Naïve Bayes -NB- algorithm for classification purposes), evolutionary optimization algorithms (e.g., the Genetic Algorithm), as well as decision-making tools (e.g., the Analytic Hierarchy Process for modeling and controlling subjective judgments) can be used in hybrid synthesis.

To begin with, the proposed framework takes advantage of the expert knowledge on the feature set that will be used for credit scoring and allows him/her to group features into homogeneous groups/clusters. Particularly, experts using the K-Means algorithm are able to create one or more clustering scenarios regarding the features by defining a desired number of clusters per scenario. For each clustering scenario, the Analytic Hierarchy Process (AHP) is applied for helping experts to make preferences for features of each cluster, set pair-wise constraints for features of equal importance, and evaluate their subjective judgments in terms of consistency. The expert input in the framework (i.e., design clustering scenarios, ranking features per each cluster per each scenario, set pair-wise constraints) using the K-Means algorithm and the AHP approach attempts to reduce the dimensionality of the dataset (i.e., decrease the number of features to be used) and define a pre-processing phase (i.e., the first phase) of the credit risk assessment process. Considering the most important features of the first phase, as well as the pair-wise constraints defined by experts, the second phase is related to a learn-heuristic algorithm that hybridize the Genetic Algorithm with supervised machine learning algorithms (i.e., the KNN algorithm and the NB algorithm) for solving the credit scoring problem. To the best of our knowledge, considering the feature selection problem in an optimization framework subject to constraints has not been investigated in the credit scoring literature.

The remainder of this paper is structured as follows. Section 2 discusses the related works. Section 3 introduces the proposed solution approach combining expert knowledge with evolutionary feature selection for credit scoring. In Section 4, the performance of the proposed solution approach in terms of numerical experiments using the German-Numeric credit dataset, which is publicly available at the UCI (University of California, Irvine) machine learning repository, is illustrated. Conclusions and future directions are given in Section 5. Finally, Appendix presents the basic concepts regarding the algorithms/methods that are used in the proposed two-phase credit risk assessment process.

2. Related work

Credit scoring has received a great deal of attention from academics, consultants and practitioners. Generally, methods for solving credit scoring problems can be divided into five groups: statistical methods, rule-based methods, multi-criteria methods, soft computing methods, and hybrid methods. It should be mentioned that works most closely related to this paper are most likely those of applying soft computing or hybrid methods. We decided to report these methods (i.e., soft computing or hybrid methods) as computational intelligence algorithms, with the term "computational intelligence" used to emphasize the application of machine learning, rule-based inference, and meta-heuristic algorithms in hybrid synthesis to solve credit scoring problems. Most recent work can be classified into two main streams, depending on whether the computational intelligence approach is combined with feature selection or not to solve credit scoring problems. Prior to analyze these computational intelligence approaches, some key points should be mentioned regarding the other credit scoring methods.

Statistical methods are related to traditional supervised machine learning methods such as the Logistic Regression (LR) algorithm and the Linear Discriminant Analysis (LDA) algorithm [35–37]. The main advantages of these two methods are that they are simple, easily understandable from business users, and easily implemented by software developers. However, many assumptions that are considered when these methods are used, such as the multivariate normality, the homoscedasticity and the absence of interactions among explanatory variables, can be seen as major weaknesses since real-word data are characterized from several data challenges and difficulties. A comprehensive review for multi-criteria and rule-based scoring methods can be found in the recent work of Soui et al. [38]. Multi-criteria methods usually reflect the process of modeling scoring problems as multi-objective optimization problems in which two or more objective functions should be optimized simultaneously (e.g., minimize the complexity of the generated solution and maximize a specific performance measure). In addition to this, multi-criteria methods such as the AHP, the ELECTRE-III, the TOPSIS, the PROMETHEE and so on can also be used for evaluating classification algorithms to predict credit risk [39,40]. The main advantage of using a multi-criteria method is that it allows researchers to model scoring problems in a more realistic context in which a trade-off between conflicting objectives should be found. With respect to the nature and the dimension of the credit scoring problem, the solution algorithms are more complex, whereas a high computation cost exists as the dimension of the problem increases. Rule-based methods are rule-extraction algorithms that can be combined with machine learning or/and evolutionary algorithms to build interpretable credit risk assessment models. Although these methods can identify patterns in complex data structures to extract rules that are easily interpretable, it is difficult to use them for large-scale credit scoring problems and extract comprehensible rules.

2.1. Computational intelligence algorithms without feature selection

As far as the methods not employing feature selection are concerned, several machine learning algorithms were proposed in the literature such as the Logistic Regression (LR), Classification and Regression Trees (CART), Neural networks (NNs) and Support Vector Machines (SVM). Baesens et al. [41] used LR and CART to build behavioral scorecards in SAS environment. Khashman [33] proposed three NN models based on the back propagation learning algorithm to decide whether to approve or reject a credit application in an automatic processing environment. A hybrid model called "boosted deep network model" combining the genetic programming meta-heuristic algorithm and the deep learning approach was proposed by Tran et al. [42]. In particular, they used genetic programming as a rule extraction engine in hybrid synthesis with the deep learning network to build a robust credit scoring model. In addition to this, Luo et al. [7] presented a deep learning approach for corporate credit scoring by applying Deep Belief Network with Restricted Boltzmann Machines. Their approach performed better than LR, Multi-layer Perceptron (MLP) and SVM in a comparison schema.

Furthermore, Moula et al. [43] mentioned that most studies on credit default prediction modeling are not directly comparable since they used different performance measures on different credit risk datasets under different circumstances. As a result, they applied SVM to solve six credit scoring problems using real-world credit risk datasets. The computational results were evaluated using ten accuracy measures (e.g., Sensitivity, Specificity, Accuracy, Area Under Curve, etc.), whereas SVM compared with five classifiers (i.e., LDA, LR, CART, MLP and Radial Basis Function - RBF) [43]. Results indicated that the SVM model was more robust than its other counterparts. For a comprehensive review on SVM-based credit scoring models through the year 2019, see [44]. An alternative approach was introduced by Soui et al. [38] who modeled the credit scoring problem as a multi-objective optimization problem to improve the interpretability and comprehensibility of rules extraction algorithms. The optimization problems solved via evolutionary optimization algorithms which attempt to maximize the rules importance and the accuracy, as well as minimize the complexity of the generated solution.

As Dastile et al. [5] mentioned in their work, three of the limitations in credit scoring literature are the usage of different cut-offs for classifying borrowers, the management of imbalanced datasets and the application of ensemble methods. In this direction, Mykola et al. [45] introduced a profit-based credit scoring approach based on the concept of reinforcement learning so as to optimize the acceptance threshold of a credit scoring model. Moreover, a reject-inference based credit scoring approach introduced by Anderson [46] who applied Bayesian Networks (BNs) to differentiate from traditional reject inference credit scoring methods, obtaining in this way better results. Recently, ensemble methods based on supervised and semi-supervised machine learning algorithms have been introduced. Xia et al. [47] proposed gradient boosting decision tree-based models (GBMs) for optimizing the ensemble selections strategies of these algorithms with respect to the overfitting measure, whereas Liu et al. [48] presented GBMs to focus on the feature augmentation process of ensemble algorithms and balance the accuracy and its interpretability. Proposed ensemble methods outperformed classic GBMs in the credit scoring literature. In addition to this, Xiao et al. [49] proposed an ensemble method based on semi-supervised machine learning algorithms to solve credit scoring problems with labeled and unlabeled samples of customers. As far as the management of imbalanced credit risk datasets is concerned, Shen et al. [50] introduced an improved synthetic minority oversampling technique (SMOTE) and solved credit scoring problems using ensemble methods based on adaptive boosting (AdaBoost) and deep learning techniques. Finally, it is worth mentioning that some researchers deal with the improvement of neural networks' performance. Extreme learning machines and neuro-fuzzy approaches introduced to propose novel activation functions for neural networks and optimize their hyper-parameters using swarm intelligence and evolutionary optimization algorithms [51–54].

2.2. Computational intelligence algorithms with feature selection

Another strand in the literature focuses on scenarios where the feature selection process is a crucial step in solving credit scoring problems. Feature selection is seen as a dimensionality reduction technique, which is indispensable due to the abundance of real-world data and the imperative to discover knowledge from raw data by selecting the most appropriate subset of features that maximizes a chosen classification measure [55,56]. Feature selection approaches can be categorized into three levels [57]. The first categorization is called "filter-based feature selection" and assumes that the features are statistically selected (ranking approach) without utilizing machine learning algorithms. The second categorization, known as the "wrapper-based feature selection", is related to an iterative scheme where a subset of features is selected and the corresponding data are used for training and testing machine learning models. A selected subset of features is evaluated based on a chosen performance classification measure, whereas the best subset of features corresponds to the iteration with the maximum value of the performance measure. Moreover, the third categorization, known as the "hybrid feature selection", is associated with the combination of filter-based and wrapper-based approaches where the first one is used to select a pool of features and then the second one is applied to find the best subset of features from this pool. It should be mentioned that due to the NP-hard nature of the feature selection problem [58] single-point search meta-heuristics such as the Tabu Search algorithm [59–61], the Greedy Randomized Adaptive Search Procedure [61] and the Variable Neighborhood Search algorithm [62], as well as population-based search meta-heuristics like genetic algorithms [61,63–66], teaching–learning based algorithms [67], gravitational search algorithms [68], and adaptive differential evolutions [69] have been introduced in the literature.

Credit scoring with hybrid-based feature selection approaches is of increasing relevance in the field or credit risk assessment [70]. A credit scoring scheme with a hybrid feature selection approach was presented by Šušteršic et al. [17]. The filter part of the model is based on Principal Component Analysis, whereas the wrapper part combines genetic algorithms (GAs) and NN. Furthermore, Chen & Li [71] proposed four feature selection approaches combined with the SVM model in credit scoring. Two of them, the LDA and the F-score, were combined with the SVM to construct credit scoring models with hybrid feature selection. Oreski & Oreski [72] combined the information gain measure (filter part) and GA with NN (wrapper part) to develop a credit scoring model for a Croatian bank. Similarly, Jadhav et al. [73] proposed an approach of combining the information gain measure for ranking the features (filter part) and GAs using three supervised machine learning algorithms (the KNN, the NB, and the SVM) as the wrapper part. A hybrid system with a filter approach (information gain, Pearson correlation coefficient and F-score) and a multiple population GA for feature selection in credit scoring was presented by Wang et al. [74]. Similarly, Trivedi [75] presented hybrid models based on feature selection techniques such as the Information-gain, the Gain-ratio and the Chi-square statistic, as well as different ML algorithms such as Random Forests and the SVM to improve the predictive performance of credit scoring. Additionally, Zhu et al. [76] proposed a hybrid-based model for consumer credit scoring combining the Relief algorithm (filter-based feature selection) and convolutional NNs. Recently, Lappas & Yannacopoulos [77] proposed an alternative hybrid model that combines the minimum redundancy maximum relevance algorithm (filter part) with a wrapper method based on GA, LR and LDA algorithms to build an accurate credit scoring model with a high predictive performance.

Credit scoring models with wrapper-based feature selection approaches have also been considered in the literature. Bellotti & Crook [78] introduced the SVM model as the basis of a feature selection method to discover the most significant subset of features in determining credit default risk. More recently, Ha & Nguyen [79] presented a wrapper-based credit scoring model based on deep learning in order to evaluate the borrower's credit score from the borrower's input features. Single-point search meta-heuristics such as the Local Search algorithm, the Stochastic Local Search algorithm and the Variable Neighborhood Search algorithm proposed to be combined with the SVM by Boughaci et al. [80] as wrapper-based feature selection approaches in credit scoring. For a comprehensive review on the use of meta-heuristics in credit scoring through the year 2019, see [44]. Finally, it is worth mentioning that several credit scoring models based on feature selection algorithms and ensemble classifiers were also introduced in the literature to improve their predictive performance [81,82].

2.3. Research motivation

The previous related works indicate that it is very difficult for someone to determine an overall "best" credit scoring model for all situations. Possible benefits and drawbacks occur when a particular credit scoring method is used. It is widely accepted that soft computing methods based on artificial intelligence and evolutionary optimization techniques perform better than other computational methods from the viewpoint of classification accuracy, but their complex structure do not facilitate (in most cases) their interpretability. On the other hand, the simplicity and assumptions that are taken into consideration in the context of traditional credit scoring methods cannot face difficulties occurred in the big (real-word) data era. Credit scoring databases are often large and characterized by redundant and irrelevant features. Solving large-scale credit scoring problems is still a challenging and difficult task. The computing complexity of feature-selection based approaches are directly affected by the credit scoring problem space. To reduce this computing complexity researchers usually promote filter-based approaches. However, when wrapper-based approaches are applied, the performance is highly improved (than in the case of filter-based approaches), but the computing complexity is increased.

To fill the gaps identified above in the credit scoring literature, we will investigate a feature selection-based credit scoring problem. Differently from all previous approaches, our method aims at combining expert knowledge with computational intelligence techniques. The ability of interpretation of the predictive power of each feature in the credit scoring dataset is strengthened by the engagement of experts in the credit scoring process, as well as the proposed wrapper process which explores how the features contributing most towards the classification of borrowers. Optimal decision making becomes a demanding task for financial analysts (i.e., experts) who play a crucial role in banks' loan approval systems in which their inputs are required for deciding whether a loan application will be approved or not. The novelty of the proposed methodology is associated with the fact that experts can express a preference for any feature of the credit scoring dataset according to their experience and set pair-wise constraints for features of equal importance. In addition to this, experts' consistency and subjective judgments are controlled. As a result, a feature-selection problem is set in an optimization framework subject to constraints so as to incorporate experts' opinions in the credit risk assessment process. Furthermore, a wrapper approach is applied to evaluate the contribution of expert opinions in the credit risk assessment process by presenting the most significant features for classifying borrowers.

It is worth mentioning that our goal is not to improve the performance of classical machine learning algorithms as it happens in many studies, but to facilitate the engagement of experts in the solution of credit scoring problems. This work focuses on integrating unsupervised ML techniques and decision-aid tools with supervised ML classifiers and evolutionary optimization algorithms at different stages, aiming to assist banks in the task

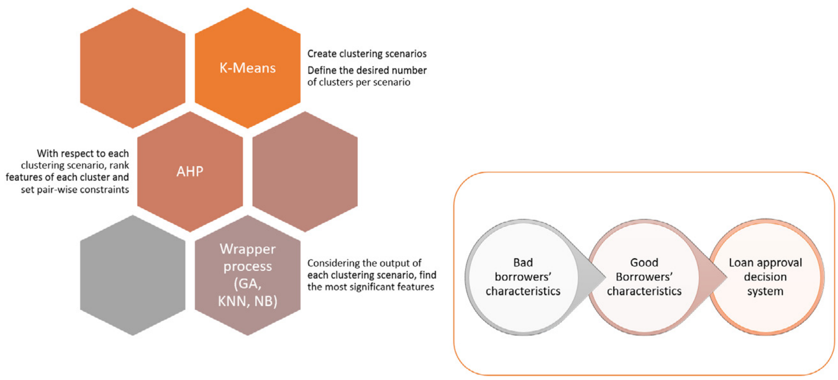

Fig. 1. Schematic diagram of the main idea of this work.

of credit risk evaluation when they use semi-automated loan approval systems and experts' inputs are required for deciding whether a loan application is approved or not. The main idea of this work is shown in Fig. 1 Initially, the K-Means algorithm (i.e., an unsupervised MLtechnique) is used to allow experts to group features into desired number of clusters (e.g., 3, 4, 5 clusters). For each cluster, an expert makes preferences to rank features based on his/her experience and sets pair-wise constraints for features of equal importance. The AHP decision making approach is used for modeling and controlling subjective judgments. Then, supervised machine learning algorithms such as the KNN and NB are combined with the GA to find the most significant features for classifying borrowers into "good" and "bad" classes. Characteristics from "good" and "bad" borrowers can be evaluated and analyzed from experts so as to design in an optimal way the existing credit risk policy and facilitate thus the decision-making approach of semi-automated loan approval systems.

Table 1 describes the credit scoring methods that were used in some correlated research papers, their performance measures and their datasets. For papers with multiple datasets and multiple performance measures, the German-numeric dataset from the UCI machine learning repository, as well as the Area Under Curve (AUC) measure are highlighted since they are also used from our paper to verify the effectiveness of the proposed methodology.

3. Methodology

The general structure of the proposed optimization framework for combining expert knowledge with evolutionary feature selection for credit risk assessment consists of two phases (Fig. 2). The basic inputs of the algorithm are (a) the credit dataset (see Section 4.2), (b) the machine learning parameters, i.e., the data splitting approach (see Section 4.4), the data transformation approach (see Section 4.3), the type of cross-validation approach (see Section 4.4), and the type of the supervised machine learning algorithm (see Sections 4.8 and 4.9), as well as (c) the genetic algorithm parameters (see Section 4.7). The outputs of the algorithm are the following: the best subset of features ever found, the machine learning model with the best parameters and the population as it evolved in the last generation of GA.

3.1. Phase I

The first phase (Phase I) represents an initial approach of dimensionality reduction where a clustering approach is used for grouping features and for each cluster, containing more than two features, the AHP decision-making methodology is applied to rank them according to the expert's (i.e., the decision maker) pair-wise comparison judgments. Based on the final ranking of features, the decision maker may set pair-wise constraints between low-score features of relatively equal importance. As a result, an optimization problem for maximizing a chosen performance measure subject to pair-wise constraints can be formulated.

Let us consider a set of N features in total, $F = \{f_1, f_2, \dots, f_N\}$, and a subset $F' \subset F$. Indicator variables x_i (binary variables), $i = 1, 2, \dots, N$, combined into a vector $x = (x_1, \dots, x_N)$, are used to indicate whether feature f_i belongs to F' or not ($x_i = 1$ if $f_i \in F'$ whereas $x_i = 0$ if $f_i \notin F'$). Moreover, let us consider a set of r pair-wise constraints defined by an expert as

$$C = \{(f_{i_1}, f_{j_1}), \dots, (f_{i_r}, f_{j_r})\} \tag{1}$$

and the corresponding set of indices

$$I_C = \{(i_1, j_1), \dots, (i_r, j_r)\} \tag{2}$$

where $i_1 \neq j_1, \dots, i_r \neq j_r$ for every $i_1, \dots, i_r, j_1, \dots, j_r \in \mathbb{N}$. The constraints consist in that the pairs of features $(f_{i_1}, f_{j_1}), \dots, (f_{i_r}, f_{j_r})$ are not assumed to offer extra information concerning the classification. Therefore, when trying to select a subset F_o of the feature set F, consisting of the minimal set of features that may be used in characterizing the credit behavior of a borrower, these features are not to be considered simultaneously into F_o.

3.2. Phase II

Phase II consists in the formulation of a constrained optimization problem, for the selection of F_o, and its solution in terms of GAs.

If $x \mapsto \Phi(x)$ represents a function which is associated with the performance measure chosen for evaluating a subset of features (i.e., AUC) via machine learning algorithms, then the feature selection problem is formulated as follows

$$\max_{x=(x_1,\dots,x_N),\in\{0,1\}^N} \Phi(x), \quad \text{s.t.} \tag{3}$$
$$x_p + x_q \leq 1, \ \ \forall (p, q) \in I_C.$$

where C and I_C are defined by Eqs. (1) and (2), respectively.

The solution $x^o = \{x_1^o, \dots, x_N^o\}$ of problem (3), will be the indicator function of the optimal feature set F_o. The second phase is associated with a wrapper-based feature selection approach where a genetic algorithm and a supervised machine learning algorithm (KNN or NB) are combined to find the subset of features F_o that maximizes a chosen performance measure.

The second phase (Phase II) starts by generating an initial population, of indicator vectors $\{x^p \in \{0, 1\}^N \ : \ p \in \{1, \dots, P\}\}$. The population initialization can be done randomly or using an opposition-based learning approach [83]. According to opposition-based learning (OBL) approach, let $x^p = (x_1^p, x_2^p, \dots, x_N^p) \in \{0, 1\}^N$,

Table 1
Related works.

Authors	Database	Computational intelligence algorithms without feature selection	Computational intelligence algorithms with feature selection	Performance measures
Huang et al. (2007)	Australian Credit Dataset, German-numeric Credit Dataset	–	SVM, Grid search, GA, F-score	Accuracy
Chen & Li (2010)	Australian Credit Dataset, German-numeric Credit Dataset	–	SVM, Rough sets, F-score, LDA, CART	Accuracy, AUC
Boughaci et al. (2018)	Australian Credit Dataset, German-numeric Credit Dataset	–	Variable Neighborhood Search, Local Search, Stochastic Local Search, SVM	Accuracy
Wang et al. (2018)	Australian Credit Dataset, German-numeric Credit Dataset	–	F-score, Information Gain, Pearson Correlation, SVM, Multiple population GA	Accuracy
Feng et al. (2019)	Australian Credit Dataset, German-numeric Credit Dataset, Loan Default Prediction Dataset	-	F-score, Pearson Correlation, Information Gain, Relief, SVM	Accuracy
Krishna & Ravi (2019)	Australian Credit Dataset, German-numeric Credit Dataset, Financial Statement Fraud Dataset	–	LR, Probabilistic NNs, NB, SVM, Adaptive Differential Evolution	Accuracy, AUC
Tripathi et al. (2019)	Australian Credit Dataset, German-numeric Credit Dataset, German-categorical Credit Dataset, Japanese Credit Dataset	–	Quadratic Discriminant Analysis, NB, Multi-layer Feed Forward NNs, time delay NN, distributed time delay NN, CART, SVM, KNN, Ensemble Classifier	Accuracy, Sensitivity, Specificity, G-measure
Xia et al. (2020)	Australian Credit Dataset, German-numeric Credit Dataset, Japanese Credit Dataset, PPDai Dataset	Random Forests, Gradient Boosting Mechanism, XGBoost, LightGBM, CatBoost	–	Accuracy, AUC, H-measure, Brier Score
Kuppili et al. (2020)	Australian Credit Dataset, German-numeric Credit Dataset, German-categorical Credit Dataset, Japanese Credit Dataset, Bankruptcy Dataset	Spiking Extreme Learning Machine	–	Accuracy
Tripathi et al. (2020)	Australian Credit Dataset, German-numeric Credit Dataset, German-categorical Credit Dataset, Japanese Credit Dataset	Evolutionary Extreme Learning Machine	–	Accuracy, Sensitivity, Specificity, G-measure, AUC
Zhang & Qiu (2020)	Australian Credit Dataset, German-numeric Credit Dataset, Japanese Credit Dataset, Taiwan Credit Dataset, Home Equity Line of Credit Dataset from US	Back-propagation Artificial NNs, Swarm Optimization Algorithm	–	Accuracy, AUC, Precision, Specificity, Sensitivity, Brier Score, G-mean
Lappas & Yannacopoulos (2021)	German-numeric Credit Dataset	–	Minimum Redundancy Maximum Relevance algorithm, LR, Linear Discriminant Analysis, GA	AUC
Liu et al. (2021)	Australian Credit Dataset, German-numeric Credit Dataset, Japanese Credit Dataset, Taiwan Credit Dataset, Lending Club Dataset, WE Dataset	Augmented Gradient Boosting Decision Trees	–	Accuracy, AUC, Brier Score, Recall Score, Precision Score
Shen et al. (2021)	German-numeric Credit Dataset, Taiwan Credit Dataset	Adaptive Boosting, Long-short-term-memory NNs	–	AUC, Kolmogorov–Smirnov Statistic

be a randomly generated indicator function corresponding to a subset of features to be evaluated. Then, for each x_i^p, $i = 1, \ldots, N$, its opposite is defined as follows $\bar{x}_i^p = 1 - x_i^p$. As a consequence, the opposite of x^p is defined as $\bar{x}^p = (1 - x_1^p, 1 - x_2^p, \ldots, 1 - x_N^p)$. For instance, if $N = 4$ and $x^p = (1, 0, 0, 1)$ then $\bar{x}^p = (0, 1, 1, 0)$. According to Rahnamayan et al. [84], the OBL approach can accelerate convergence speed of evolutionary algorithms and as a result is tested in the wrapper approach of the Phase II. Feasible individuals (i.e., those that do not violate constraints) are kept for constructing the initial population (i.e., "Pop") based on a predefined population size. For each individual of the initial population (i.e., a selected subset of features) a machine learning (ML) algorithm is created (Fig. 3). In particular, the best parameters of ML algorithm obtained via a k-fold cross validation process where the already transformed training set is divided into training and validation sets (see Section 4.4). The newly trained ML algorithm

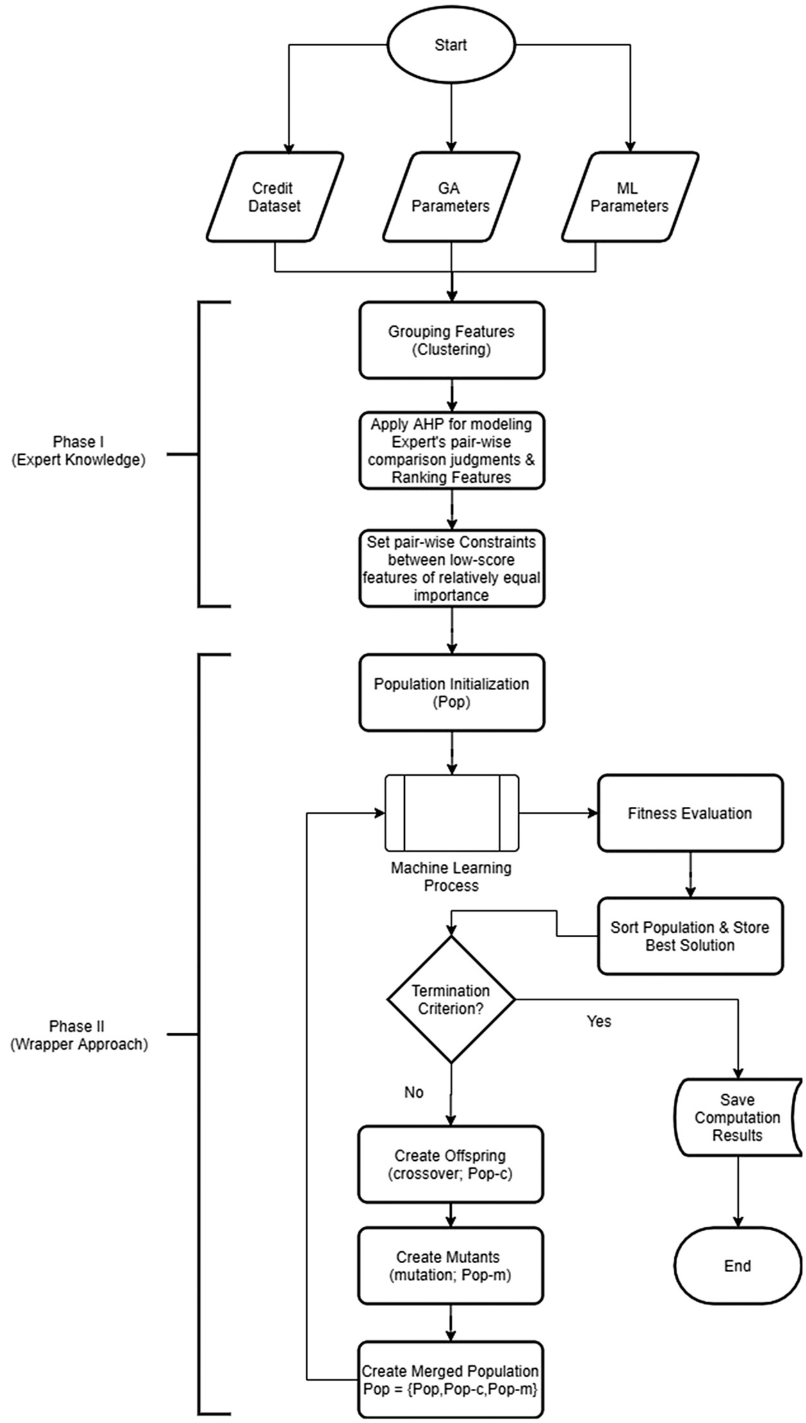

Fig. 2. Flowchart of the proposed solution approach.

is evaluated on Test Set with a chosen performance measure (i.e., AUC).

After evaluating all the individuals of the population, "Pop" is sorted. According to the population size, and the predefined crossover and mutation percentages, the number of offspring (i.e., "Pop-c"), as well as the number of mutants (i.e., "Pop-m") can be computed, respectively. The crossover process assumes that two parents are chosen using a parent selection method (see Section 4.7). Since parents are selected, a crossover operator is applied to generate at least one new offspring. This process is repeated until the appropriate number of offspring is reached (i.e., "Pop-c"). Furthermore, a parent is randomly selected for applying a mutation (see Section 4.7). The mutation process is repeated until the appropriate number of mutants is reached (i.e., "Pop-m"). "Pop-c" and "Pop-m" are then evaluated and merged with "Pop". The new population is sorted and the best solution is saved. Based on the population size, the number of best individuals to be kept for the next generation is equal to the predefined size of the population. The GA is repeated until a termination criterion is satisfied (see Section 4.7).

In the context of the proposed optimization framework, two parent selection methods, one crossover operator type and one mutation operator type are supported. As far as the parent selection phase is concerned, the roulette-wheel selection (RWS) and

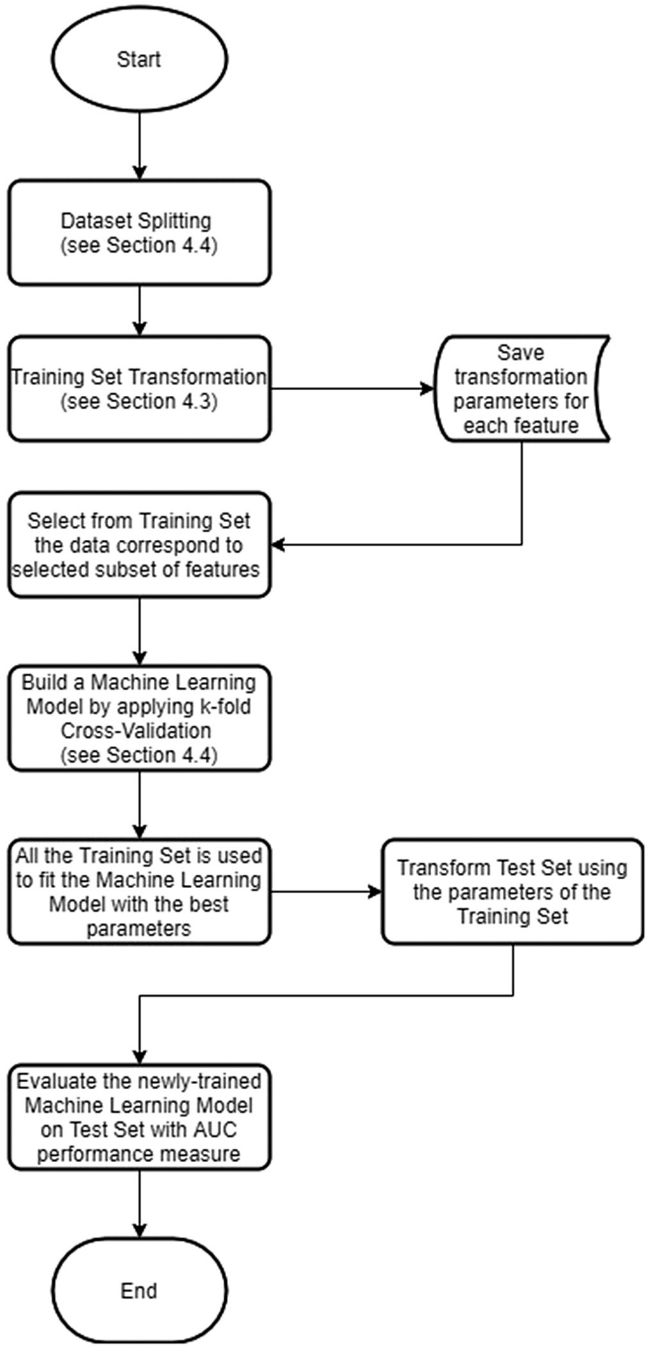

Fig. 3. Flowchart for the Machine Learning Process.

the stochastic universal sampling (SUS) are implemented [85]. The RWS assigns to each parent within a population a selection probability that is proportional to its relative fitness value. As a result, better parents have higher probability to be selected for the crossover phase. The SUS maps individuals to contiguous segments of a line, such that each individual's segment is equal in size to its fitness value, as in RWS. Then, equally spaces pointers are placed over the line as many as there are individuals to be selected. Considering as *nPointer* the number of individuals to be selected, then the distance between the pointers are $\frac{1}{nPointer}$ where the position of the first pointer is given by a randomly generated number in the range $[0, \frac{1}{nPointer}]$.

As far as the crossover operator is concerned, the concept of the Subset Size-Oriented Common Features Crossover Operator (SSOCF) presented by Emmanouilidis et al. [86] is used. In particular, crossover operator combines genes of two parents and produces at least one offspring ($\leq$ 2). Two parents are initially scanned for common genes. The common genes are inherited to both offspring. Then, the remaining non-common genes from both parents are randomly shared between two offspring. The two offspring are evaluated in terms of constraints and if at least one of them does not violate constraints it is added to the population (i.e., "Pop-c"). If both offspring violate constraints, the parent selection and the crossover phases are repeated.

It should be mentioned that the mutation phase is highly influenced by a parameter called "mutation rate" (see Section 4.7). Actually, the mutation rate influences the number of genes (i.e., features) of a randomly selected parent-chromosome that will be replaced with new ones. Based on the length of a chromosome, the number of genes is usually one or two. In this paper, the mutation operator is a heuristic approach by which the selected features to be replaced are those involved in the largest number of violated constraints, and the new features are those which maximize the number of constraints that become satisfied.

3.3. Time complexity

It is well known that (in most cases) soft computing approaches are associated with approximation algorithms that aim at solving Nondeterministic-Polynomial (NP)-hard problems [87, 88], namely problems from which a solution cannot be obtained in a polynomial time. The complexity of these problems can be described in terms of time (i.e., how long a computation takes to execute) or/and in terms of space (i.e., how much storage is required for a computation). An interesting approach is to analyze the space complexity in combination with time complexity when a cluster of computers/servers can be available via the cloud and parallel computing concepts are applied. The credit scoring problem in the form of a feature selection-based optimization problem subject to constraints is an NP-Hard problem [58]. As the main goal of this paper is to design, develop, and evaluate approximation algorithms for the credit scoring problem in a single machine, the time complexity of the problem is briefly described with respect to the selected computational algorithms.

The computational effort that is required for solving the credit scoring problem can be measured by the number of arithmetic or logical operations with respect to the total number of iterations and so on. Depending on the scope and size of the problem, the big O mathematical notation can be used for describing the limiting behavior of an evaluation function when the corresponding arguments tends towards a particular value or infinity. The time complexity of the K-means algorithm is $O(LKN)$, where N is the number of features, K is the desired number of groups, and L is the number of loops needed to complete the clustering process [89]. With the application of the genetic algorithm the time complexity becomes $O(Mlog(M) + MNPG)$, where M is the number of borrowers, N is the number of features, P is the population size, and G is the number of generations [90]. As the values of the aforementioned arguments are increased, more CPU time and more memory are needed for the execution.

The concept of the time complexity becomes more interesting when we focus on the fitness evaluation function of the GA. The fitness value obtained after applying supervised machine learning algorithms. These algorithms are trained via a cross-validation process, whereas the fitness value obtained in the inference phase (i.e., test phase). As a result, time complexity should be evaluated in both phases. The time complexity in the training phase for the NB and KNN algorithms is $O(TN)$, where T is the number of training examples, and N is the number of features. In the inference phase, the time complexity of the algorithms is $O(JN)$ since we have to retrieve N feature values for each of the J classes (i.e., good/bad borrowers).

The time complexity is also associated with the application of the Analytic Hierarchy Process. According to Tung [91], the time complexity of the AHP method can be seen per each level of the AHP architecture (for more details related to the AHP method see Appendix A.2). In this work, the AHP architecture consists of a single level. In particular, the level includes the features of a specific cluster for which an expert makes his/her preferences. As a result, the AHP approach is repeated for each cluster that contains more than one features. In addition to this, the overall process can be repeated for any desired clustering scenario of the expert (e.g., group features into 4 clusters or group features into 8 clusters, etc.). Assuming that the single level of the AHP architecture has R features, an RxR matrix should be filled with experts' preferences. The time complexities for deriving the largest eigenvalue and solving the simultaneous equations at this single level of the AHP architecture is $O(min(RC^2, R^2C))$, where R represents features and C represents criteria [92].

As it can be observed the time complexity of our solution algorithms is affected in many ways due to the number of parameters. Therefore, a trade-off between the computational complexity and the effectiveness of the algorithms should be investigated. The main goal of the paper is not to compute the highest accuracy or the highest AUC in a specific credit scoring dataset of the literature. We are of the opinion that soft computing algorithms aim at finding near-optimal solutions for real-world data-driven problems. Depending on the available data, specific data management and computational intelligence strategies should be investigated to maximize the success ratio of recognizing good/bad borrowers in banking environments. Only a (real-world) data-driven soft computing algorithm that is running in a production environment can bring value. Our main goal is to facilitate the engagement of experts in a semi-automated loan approval process in which their opinions are required from a bank institute. Considering that different experts may have different opinions in the same problem based on their experience, our approach aims at modeling, controlling and measuring subjective judgments in terms of consistency. In practice, when a credit policy is designed for a bank, experts' role is very crucial due to their experience.

The basic idea is to take advantage of the expert knowledge on the feature set of credit scoring dataset to improve the performance of the computational intelligence algorithms by providing them with good starting points. To some degree, experts' inputs can reduce the complexity of the problem. Furthermore, we try to manage the time-complexity of the algorithms using feature selection mechanisms and not considering all the features of the dataset. The parameters of GA and ML algorithms have been initially selected according to some rule-of-thumbs in the literature of evolutionary learning and machine learning. For instance, a good rule of thumb is to use $5 - 30\%$ of the number of features as a population size. Additionally, general guidelines suggest value of crossover percentage no less than 60%, and value of mutation rate equal to $1/lc$, where lc is the length of the chromosome. Generally, the number of neighbors for the KNN algorithm can be

Table 2
Credit dataset.

Feature	Description
F1	Status of existing checking account
F2	Duration in months
F3	Credit history
F4	Credit amount
F5	Savings balance
F6	Present employment since
F7	Personal status and sex
F8	Present residence since
F9	Property
F10	Age in years
F11	Other installment plans
F12	Number of existing credits at this bank
F13	Number of people being liable to provide maintenance for
F14	Telephone
F15	Foreign worker
F16	Purpose – new car
F17	Purpose – old car
F18	Other debtor – none (compared to guarantor)
F19	Other debtor – co-applicant (compared to guarantor)
F20	Housing – rent (compared to “for free”)
F21	Housing – own (compared to “for free”)
F22	Job – unemployed (compared to management)
F23	Job – unskilled (compared to management)
F24	Job – skilled (compared to management)

Table 3
Clustering scenarios.

Scenario	Feature allocation
2-means	Cluster 1: F2,F4,F10
	Cluster 2: F1,F3,F5,F6,F7,F8,F9,F11,F12,F13,F14,F15,F16,F17,F18, F19,F20,F21,F22,F23,F24
3-means	Cluster 1: F1,F3,F5,F6,F7,F8,F9,F11,F12,F13,F14,F15,F16,F17,F18, F19,F20,F21,F22,F23,F24
	Cluster 2: F4
	Cluster 3: F2,F10
4-means	Cluster 1: F4
	Cluster 2: F1,F3,F5,F6,F7,F8,F9,F11,F12,F13,F14,F15,F16,F17,F18,F19, F20,F21,F22,F23,F24
	Cluster 3: F10
	Cluster 4: F2
5-means	Cluster 1: F12,F13,F14,F15,F16,F17,F18,F19,F20,F21,F22,F23,F24
	Cluster 2: F10
	Cluster 3: F2
	Cluster 4: F4
	Cluster 5: F1,F3,F5,F6,F7,F8,F9,F11
6-means	Cluster 1: F12,F13,F14,F15,F16,F17,F18,F19,F20,F21,F22,F23,F24
	Cluster 2: F10
	Cluster 3: F4
	Cluster 4: F2
	Cluster 5: F6,F8,F9
	Cluster 6: F1,F3,F5,F7,F11
7-means	Cluster 1: F16,F17,F19,F20,F21,F22,F23,F24
	Cluster 2: F2
	Cluster 3: F4
	Cluster 4: F6
	Cluster 5: F10
	Cluster 6: F5,F12,F13,F14,F15,F18
	Cluster 7: F1,F3,F7,F8,F9,F11
8-means	Cluster 1: F6,F8
	Cluster 2: F10
	Cluster 3: F2
	Cluster 4: F4
	Cluster 5: F16,F17,F19,F20,F22,F23,F24
	Cluster 6: F12,F13,F14,F15,F18,F21
	Cluster 7: F1,F5,F7,F9,F11
	Cluster 8: F3
9-means	Cluster 1: F12,F13,F14,F15,F18
	Cluster 2: F4
	Cluster 3: F10
	Cluster 4: F2
	Cluster 5: F16,F17,F19,F20,F21,F22,F23,F24
	Cluster 6: F5
	Cluster 7: F7,F9
	Cluster 8: F6,F8,F11
	Cluster 9: F1,F3
10-means	Cluster 1: F3
	Cluster 2: F4
	Cluster 3: F12,F13,F14,F15,F18,F21,F24
	Cluster 4: F2
	Cluster 5: F10
	Cluster 6: F5
	Cluster 7: F16,F17,F19,F20,F22,F23
	Cluster 8: F6
	Cluster 9: F9
	Cluster 10: F1,F7,F8,F11

chosen to be the square root of M, where M is the number of borrowers. After running some experiments of the genetic algorithm, we tuned the values of the parameters so as the computational algorithms can provide experts with results in a reasonable time. The parameters for the GA and ML algorithms are presented in Section 4. The tables with computational results present also the computation time needed to obtain a solution.

4. Computational experiments and results

4.1. Experimental setup

The model proposed in Section 3 was developed in the MATLAB programming language (R2016a) and executed on a DELL personal computer with an Intel(R) CoreTM i3-2120, clocked at 3.30 GHz, a microprocessor with 4 GB of RAM memory under the operating system Microsoft Windows 7 professional.

4.2. Dataset

It should be mentioned that the main purpose of this paper is to describe the methodology that has been developed and its methodological parts based on machine learning and evolutionary learning approaches. Although different datasets could be potential to be selected from the literature, the selected dataset which consists of only quantitative features is used as an illustrative example since it is publicly available and commonly usable in the literature. In this direction, it can also be used for comparison purposes (see Tables 16 and 17). Particularly, the "german.data-numeric" credit dataset from the UCI Machine

Learning Repository is used to evaluate the performance of the proposed algorithms. The features of the dataset are presented in Table 2.

The dataset contains instances for 1000 past credit applicants on 24 numeric features. 700 instances are related to the good class (i.e., the value of the dependent variable is equal to 1), whereas 300 instances are associated with the bad class (i.e., the value of the dependent variable is equal to 2).

4.3. Data transformation

In this paper, the dataset is standardized to have mean 0 and standard deviation 1. The standardized dataset retains the shape properties of the original dataset [93,94]. The approach is also known as z-score method. In particular, for each feature with mean $\overline{X}$ and standard deviation S, the z-score of data point x is $z = \frac{x-\overline{X}}{S}$. It should be mentioned that since a dataset is divided into training and test sets (see Section 4.4), the data transformation process takes place as follows: (a) standardize the training set, (b) keep the parameters of transformation (i.e., mean and standard deviation for each feature) and (c) transform the test set using the parameters of the training set.

4.4. Data splitting

Three sets of data are generally used: the training set, the validation set and the test set [93,94]. The initial dataset is divided into training and test sets, with an 80-20 split (i.e., 80% of the data set to train a machine learning model and 20% to test the trained model). The training set is further divided into training and validation sets to keep track of how well a machine learning model is doing as it learns in the context of a k-fold cross validation process.

In this paper, a Stratified k-fold cross validation with $k = 10$ is used to divide training set into 10 equal-sized parts, of which, one part is used as a validation set, and nine parts as training sets. The results of the 10 iterations are averaged. Actually, cross-validation is called Stratified due to the fact that the process of splitting the training data into folds ensure that each fold has the same proportion of the two types of class labels (i.e., good/bad borrowers). It should be mentioned that cross-validation is used for tuning the parameters of a machine learning model. Finding the best parameters of a machine learning model, all the training set is used to fit that model.

4.5. Clustering

In the context of the pre-processing phase (i.e., Phase I), an expert can define several scenarios deciding the number of clusters into which features have to be grouped (i.e., number of clusters per scenario). By allowing an expert to decide the number of clusters per scenario, we would like to mention the significance of his/her opinion in the decision making process. In case of large set of features, an expert can examine ways of grouping the features into several clusters so as to achieve clusters with manageable number of features to be evaluated. It should be mentioned that for each scenario, the wrapper-based feature selection approach (i.e., Phase II) is applied to find the optimal solution.

Therefore, the k-means algorithm is used for grouping features of the original dataset into predefined number of clusters. Since clusters are going to be used from a decision maker to evaluate features per cluster, the k-means algorithm is executed 9 times to find the optimal allocation of features in 2, 3, 4, 5, 6, 7, 8, 9 and 10 clusters, respectively (Table 3). It should be noticed that the clusters presented in Table 3 for each scenario are obtained after independent executions of the k-means algorithm. For instance, there is no relationship between "Cluster 2" of scenario "2-means" and "Cluster 2" of scenario "3-means" and so on.

Table 4
Pair-wise comparison matrix (cluster 5; 8-means algorithm).

	F16	F17	F19	F20	F22	F23	F24
F16	1	2	1/3	3	1/3	1/3	1/4
F17	1/2	1	1/3	2	1/2	1/2	1/3
F19	3	3	1	3	2	2	1/3
F20	1/3	1/2	1/3	1	1	1	1/3
F22	3	2	1/2	1	1	1	1/4
F23	3	2	1/2	1	1	1	1/4
F24	4	3	3	3	4	4	1

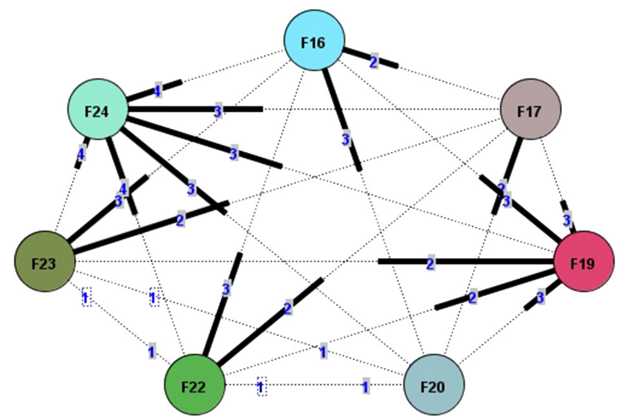

Fig. 4. Graph view for intransitive set of judgments (with reference to Table 5).

4.6. AHP

According to the scenarios presented above, the AHP method is applied to model decision maker's pair-wise comparison judgments for clusters that contain more than 2 features. The basic goal here is to rank features within a cluster, select the most significant of them (based on ranking) and set pair-wise constraints between low-score features of relatively equal importance (i.e., an initial approach of dimensionality reduction). The decision making process with AHP have been implemented using the PriEsT (stands for Priority Estimation Tool) decision support tool that developed by Siraj et al. [95].

An illustrative example is used to demonstrate key modeling features of the AHP using PriEsT. Tables 4 and 5 present the pair-wise comparison matrices related to cluster 5 of scenario with 8-means algorithm and cluster 1 of scenario with 6-means algorithm, respectively. In addition to this, Figs. 4 and 5 depict the graph views for intransitive set of judgments corresponding to Tables 4 and 5, respectively.

The priority vectors for each scenario and for each cluster, the consistency ratio (CR) of each pair-wise comparison matrix, as well as the pair-wise constraints obtained per scenario are presented in Table 6. It should be noticed that scenarios 2, 3 and 4 are merged to one scenario called "(2–4) -means scenario" since the allocation of features for each scenario is lending to the same pair-wise constraints (there is only one basic cluster including more than 2 features and it is the same to the three scenarios).

4.7. GA parameter selection

Four wrapper approaches are used in this paper, namely "defaultGA+KNN", "altGA+KNN", "defaultGA+NB" and "altGA+NB". The parameter setting for KNN and NB are the same in the above approaches as they are presented in Sections 4.8 and 4.9, respectively. The main difference between "defaultGA" and "altGA" concepts are related to the selected parameters regarding: (a) the population initialization method and (b) the parent selection

Table 5
Pair-wise comparison matrix (cluster 1; 6-means algorithm).

	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24
F12	1	2	4	3	3	3	3	1/2	3	2	3	3	1/3
F13	1/2	1	4	3	2	2	2	1/2	1/2	1/2	2	3	1/3
F14	1/4	1/4	1	1/2	1/2	1/2	1/2	1/4	1/2	1/3	1/2	1/2	1/4
F15	1/3	1/3	2	1	1/2	1/2	1	1/3	2	1/2	2	3	1/2
F16	1/3	1/2	2	2	1	2	2	1/3	2	2	2	2	1/3
F17	1/3	1/2	2	2	1/2	1	2	1/3	2	2	2	2	1/3
F18	1/3	1/2	2	1/2	1/2	1/2	1	1/3	1/2	1/2	1/2	1	1/4
F19	2	2	4	3	3	3	3	1	2	3	3	3	1/2
F20	1/3	2	2	1/2	1/2	1/2	2	1/2	1	1/2	2	2	1/3
F21	1/2	2	3	2	1/2	1/2	2	1/3	2	1	3	3	1/2
F22	1/3	1/2	2	1/2	1/2	1/2	2	1/3	1/2	1/3	1	2	1/3
F23	1/3	1/3	2	1/3	1/2	1/2	1	1/3	1/2	1/3	1/2	1	1/4
F24	3	3	4	2	3	3	4	2	3	2	3	4	1

Table 6
Priority vectors and pair-wise constraints.

	Priority vector per cluster & pair-wise constraints
(2-4)-means scenario	Basic cluster weights: 0.066 0.062 0.061 0.035 0.028 0.028 0.038 0.101 0.076 0.047 0.013 0.026 0.043 0.036 0.025 0.101 0.028 0.043 0.023 0.02 0.1 Constraints: F6,F17; F9,F14; F16,F21; F7,F8; F22,F23; F15,F18 / CR: 0.09 (9% <10%)
5-means scenario	Cluster 1 weights: 0.128 0.08 0.027 0.056 0.074 0.067 0.036 0.147 0.056 0.078 0.042 0.033 0.177 Constraints: F14,F23; F18,F22; F15,F20 / CR: 0.052 (5.2% <10%) Cluster 5 weights: 0.203 0.229 0.136 0.071 0.068 0.046 0.072 0.174 Constraints: F6,F8; F7,F9 / CR: 0.064 (6.4% <10%)
6-means scenario	Cluster 1 weights: 0.128 0.08 0.027 0.056 0.074 0.067 0.036 0.147 0.056 0.078 0.042 0.033 0.177 Constraints: F14,F23; F18,F22; F15,F20 / CR: 0.052 (5.2% <10%) Cluster 5 weights: 0.594 0.157 0.249 Constraints: No / CR: 0.046 (4.6% <10%) Cluster 6 weights: 0.246 0.232 0.141 0.099 0.282 Constraints: F5,F7 / CR: 0.050 (5% <10%)
7-means scenario	Cluster 1 weights: 0.068 0.074 0.192 0.06 0.105 0.095 0.095 0.311 Constraints: F16,F17; F16,F20; F22,F23 / CR: 0.092 (9.2% <10%) Cluster 6 weights: 0.228 0.323 0.19 0.055 0.087 0.117 Constraints: F14,F15 / CR: 0.027 (2.7% <10%) Cluster 7 weights: 0.246 0.29 0.103 0.062 0.084 0.215 Constraints: F8,F9 / CR: 0.083 (8.3% <10%)
8-means scenario	Cluster 5 weights: 0.08 0.072 0.201 0.068 0.115 0.115 0.349 Constraints: F16,F17; F17,F20 / CR: 0.085 (8.5% <10%) Cluster 6 weights: 0.354 0.255 0.06 0.085 0.102 0.144 Constraints: F14,F15 / CR: 0.041 (4.1% <10%) Cluster 7 weights: 0.173 0.375 0.074 0.131 0.247 Constraints: F7,F9 / CR: 0,067 (6.7% <10%)
9-means scenario	Cluster 1 weights: 0.365 0.373 0.043 0.068 0.152 Constraints: F14,F15 / CR: 0.061 (6.1% <10%) Cluster 5 weights: 0.068 0.074 0.192 0.06 0.105 0.095 0.095 0.311 Constraints: F16,F17; F16,F20; F22,F23 / CR: 0.092 (9.2% <10%) Cluster 8 weights: 0.268 0.117 0.614 Constraints: No / CR: 0.063 (6.3% <10%)
10-means scenario	Cluster 3 weights: 0.311 0.306 0.027 0.037 0.143 0.078 0.098 Constraints: F14,F15; F21,F24 / CR: 0.099 (9.9% <10%) Cluster 7 weights: 0.107 0.085 0.254 0.077 0.211 0.266 Constraints: F17,F20 / CR: 0.059 (5.9% <10%) Cluster 10 weights: 0.453 0.167 0.118 0.262 Constraints: No / CR: 0.026 (2.6% <10%)

Table 7
GA parameters.

Parameter	Value
Fitness Function	AUC value
Population Size	25-30
Number of Generations	20-50
Crossover Percentage	0.8
Mutation Percentage	Computed as the 1/(length of chromosome)
Mutation Rate	0.1-0.2
Termination Criterion	25-50 generations

method. Under the "defaultGA" concept the RWS is used as a parent selection method, while the population is randomly generated. However, in the context of the "altGA" concept, the initial population is generated by applying the OBL approach, while the SUS parent selection method is used. In addition to this, Table 7 presents the rest (common) values of the GA parameters.

4.8. KNN parameter selection

In the context of the KNN algorithm, the distance metric is set by default to the Euclidean distance, while the number of neighbors is tuned using the 10-fold cross validation process. All

Table 8
Computational Results – "2-4 means scenario".

Wrapper	Best AUC	BEST GC	Worst GC	KNN parameter	NB parameter	Number of features	Computation time (HH:MM:SS)
defaultGA+KNN Selected Features F3,F4,F14,F21,F2, F5,F6,F1	0.7895	0.5791	0.0940	19 neighbors	–	8	01:06:40
altGA+KNN Selected features F3,F4,F12,F13,F18,F10, F5,F16,F19, F20,F23,F6, F9,F1	0.7534	0.5069	0.1908	21 neighbors	–	14	01:34:08
defaultGA+NB Selected features F3,F4,F12, F15,F24,F2, F16,F17,F20, F22,F9,F1	0.7660	0.5319	0.2135	–	Triangular kernel	12	01:46:42
altGA+NB Selected features F3,F4,F12,F15,F2,F10, F5,F16,F6,F9,F1,F8,F11	0.7429	0.4857	0.2243	–	Gaussian kernel	13	01:58:00

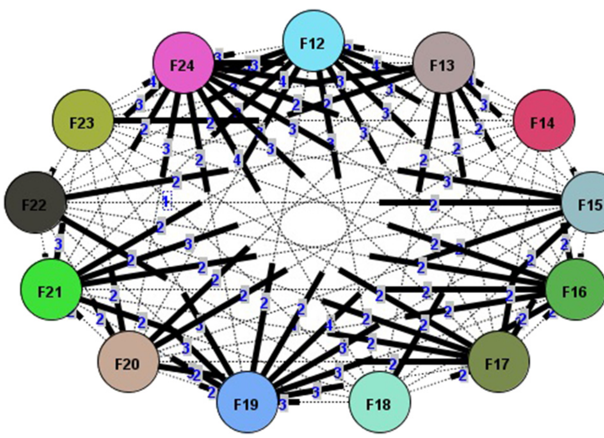

Fig. 5. Graph view for intransitive set of judgments (with reference to Table 6).

odd values of k from 1 to 21 are tried to find the optimal number of neighbors which best generalizes the data.

4.9. NB parameter selection

Regarding the NB algorithm, the kernel type parameter is tuned using the 10-fold cross validation process [96]. Kernel can be (a) Uniform, (b) Epanechnikov, (c) Gaussian and (d) Triangular.

4.10. Results

For each scenario presented in Section 4.6, the four wrapper approaches described in Section 4.7 are applied (28 independent execution runs) to illustrate the performance of the solution approaches in terms of numerical experiments using the "german.data-numeric" credit dataset from the UCI Machine Learning Repository (see Section 4.2). The results obtained with each of the proposed approaches are shown in Tables 8–14. According to Breiman et al. [97], AUC is closely related to the Gini Coefficient (GC) which is defined as $2AUC-1$. Each table presents, per execution of a scenario, the best value obtained for the AUC performance measure, the best GC, the worst GC, the optimal parameter for the selected machine learning algorithm, the number of selected features, selected features, as well as the computation time (in HH:MM:SS format, namely hours:minutes:seconds) needed to obtain the solution.

As it can be observed, the "altGA+NB" approach is more accurate in case of scenarios "7-means scenario", "8-means scenario" and "9-means scenario" than the other three wrapper approaches. The AUC value obtained in each scenario is 77.01% (using Gaussian kernel and 10 features), 76.54% (using Triangular kernel and 5 features) and 77.63% (using Triangular kernel and 7 features), respectively. Moreover, the "defaultGA+KNN" wrapper approach outperforms the other three wrapper approaches in case of scenarios "2-4 means scenario", "6-means scenario", and '10-means scenario", obtaining AUC values of 78.95% (using 19 neighbors and 8 features), 75.49% (using 21 neighbors and 10 features) and 79.70% (using 21 neighbors and 9 features), respectively. In addition to this, the "defaultGA+NB" wrapper approach seems to be better than the other three approaches in case of scenario "5-means scenario", obtaining 76.48% AUC value (using Triangular kernel and 15 features). Fig. 6 shows the ROC graph and the solution progress with respect to the fitness value (i.e., AUC) for scenario "10-means scenario" which is associated with best AUC value obtained (79.70%), using the "defaultGA+KNN" wrapper approach. As far as the computation time is concerned, it is worth to notice that our soft computing algorithms tried to find a near-optimal solutions for each scenario within a reasonable time (1-2 h).

Furthermore, Table 15 presents some summary statistics (minimum: Min, maximum: Max, average: Avg, standard deviation: Std) related to the AUC and the number of features (# Fs). Actually, a focus is placed on (a) the combination of the GA with the KNN algorithm ("GA+KNN"; 14 execution runs) and (b) the combination of the GA with the NB algorithm ("GA+NB"; 14 execution runs) to solve the credit scoring problem (i.e., the "german.data-numeric" credit dataset), irrespective of scenarios obtained from Phase I. Based on the average AUC, the "GA+KNN" approach of "german.data-numeric" credit dataset is the largest one.

Finally, the performances of "GA+KNN" and "GA+NB" are compared with the approaches proposed by Chen & Li [71], Krishna & Ravi [69], Lappas & Yannacopoulos [77], Zhang & Qiu [54], Tripathi et al. [53], Xia et al. [47], Liu et al. [48], and Shen et al. [50]. As shown in Tables 16 and 17, both proposed wrapper-based approaches (i.e., "GA+KNN" and "GA+NB") improve in some cases the results and perform well as compared to reference algorithms. The results indicate that measuring the consistency of experts, their opinions may contribute to reach near-optimal solutions in a reasonable time.

Table 9
Computational Results – "5 - means scenario".

Wrapper	Best AUC	BEST GC	Worst GC	KNN parameter	NB parameter	Number of features	Computation time (HH:MM:SS)
defaultGA+KNN Selected Features F3,F4,F13,F2,F17,F20, F9,F1,F8	0.7577	0.5153	0.1626	17 neighbors	–	9	01:10:09
altGA+KNN Selected features F3,F4,F13,F15,F24,F2, F17,F19,F1,F7,F11 F9,F1	0.7575	0.5150	0.1457	15 neighbors	–	11	00:57:18
defaultGA+NB Selected features F12,F13,F14,F15,F18,F21, F2,F10,F5,F17,F19,F1, F7,F8,F11	0.7648	0.5295	0.3535	–	Triangular kernel	15	02:12:13
altGA+NB Selected features F3,F12,F13,F15,F10,F5, F16,F17,F22,F23,F9,F1, F8,F11	0.7365	0.4729	0.2270	–	Uniform kernel	14	02:04:02

Table 10
Computational Results – "6 - means scenario".

Wrapper	Best AUC	BEST GC	Worst GC	KNN parameter	NB parameter	Number of features	Computation time (HH:MM:SS)
defaultGA+KNN Selected Features F3,F4,F13,F18,F21,F2, F5,F20,F1,F8	0.7549	0.5097	0.0679	21 neighbors	–	10	01:15:41
altGA+KNN Selected features F3,F4,F13,F2,F1	0.7391	0.4783	−0.0036	13 neighbors	–	5	00:58:31
defaultGA+NB Selected features F3,F15,F21,F16,F17,F9, F1,F11	0.7465	0.4931	0.0683	–	Triangular kernel	8	02:29:12
altGA+NB Selected features F3,F21,F24,F2,F5,F16, F17,F6,F9,F1,F11	0.7489	0.4979	0.1519	–	Triangular kernel	11	01:44:36

Table 11
Computational Results – "7 - means scenario".

Wrapper	Best AUC	BEST GC	Worst GC	KNN parameter	NB parameter	Number of features	Computation time (HH:MM:SS)
defaultGA+KNN Selected Features F3,F4,F21,F2,F10,F16, F6,F1,F7,F11	0.7613	0.5226	0.1594	21 neighbors	–	10	01:01:23
altGA+KNN Selected features F3,F4,F12,F18,F5,F17, F23,F1,F11	0.7505	0.5010	0.0642	21 neighbors	–	9	01:13:24
defaultGA+NB Selected features F3,F4,F14,F18,F2,F5, F16,F19,F23,F6,F1,F7, F11	0.7606	0.5212	0.2997	–	Gaussian kernel	13	01:54:41
altGA+NB Selected features F3,F12,F18,F21,F24,F16, F19,F6,F1,F11	0.7701	0.5402	0.1457	–	Gaussian kernel	10	01:29:50

Table 12
Computational Results – "8 - means scenario".

Wrapper	Best AUC	BEST GC	Worst GC	KNN parameter	NB parameter	Number of features	Computation time (HH:MM:SS)
defaultGA+KNN Selected Features F3,F24,F5,F1,F11	0.7573	0.5145	−0.0570	21 neighbors	–	5	00:41:23
altGA+KNN Selected features F3,F4,F12,F18,F21,F10, F5,F17,F6,F1,F7,F11	0.7520	0.5040	0.0633	21 neighbors	–	12	01:29:07
defaultGA+NB Selected features F3,F4,F14,F5,F19,F20, F23,F6,F1	0.7425	0.4850	0.1844	–	Gaussian kernel	9	01:14:42
altGA+NB Selected features F15,F24,F22,F9,F1	0.7654	0.5307	−0.0073	–	Triangular kernel	5	00:52:06

Table 13
Computational Results – "9 - means scenario".

Wrapper	Best AUC	BEST GC	Worst GC	KNN parameter	NB parameter	Number of features	Computation time (HH:MM:SS)
defaultGA+KNN Selected Features F3,F21,F24,F2,F10,F17, F19,F23,F6,F9,F1,F11	0.7551	0.5102	0.1481	21 neighbors	–	12	01:16:48
altGA+KNN Selected features F4,F18,F2,F17,F6,F1, F11	0.7717	0.5434	−0.0388	21 neighbors	–	7	00:45:20
defaultGA+NB Selected features F3,F15,F21,F2,F5,F16, F19,F6,F1,F7	0.7625	0.5250	0.1143	–	Triangular kernel	10	01:48:21
altGA+NB Selected features F3,F15,F17,F20,F1,F7, F11	0.7763	0.5525	0.1630	–	Triangular kernel	7	01:02:08

Table 14
Computational Results – "10 - means scenario".

Wrapper	Best AUC	BEST GC	Worst GC	KNN parameter	NB parameter	Number of features	Computation time (HH:MM:SS)
defaultGA+KNN Selected Features F4,F12,F18,F2,F10,F5, F22,F9,F1	0.7970	0.5940	0.0130	21 neighbors	–	9	01:36:37
altGA+KNN Selected features F3,F4,F13,F18,F2,F5, F16,F20,F6,F9,F1,F7, F11	0.7647	0.5294	0.1090	19 neighbors	–	13	01:06:32
defaultGA+NB Selected features F4,F15,F2,F16,F23,F6, F9,F1,F7,F11	0.7564	0.5129	0.2047	–	Gaussian kernel	10	01:07:27
altGA+NB Selected features F3,F4,F2,F5,F16,F20, F23,F6,F1,F7,F11	0.7565	0.5130	0.1934	–	Gaussian kernel	11	01:31:01

Table 15
Summary Statistics for each wrapper-based combination.

	Min AUC	Max AUC	Avg AUC	Std AUC	Min # Fs	Max # Fs	Avg # Fs	Std # Fs
GA+KNN	0.7391	0.7970	0.7616	0.0154	5	15	10	3.0849
GA+NB	0.7365	0.7763	0.7569	0.0118	5	15	11	2.7656

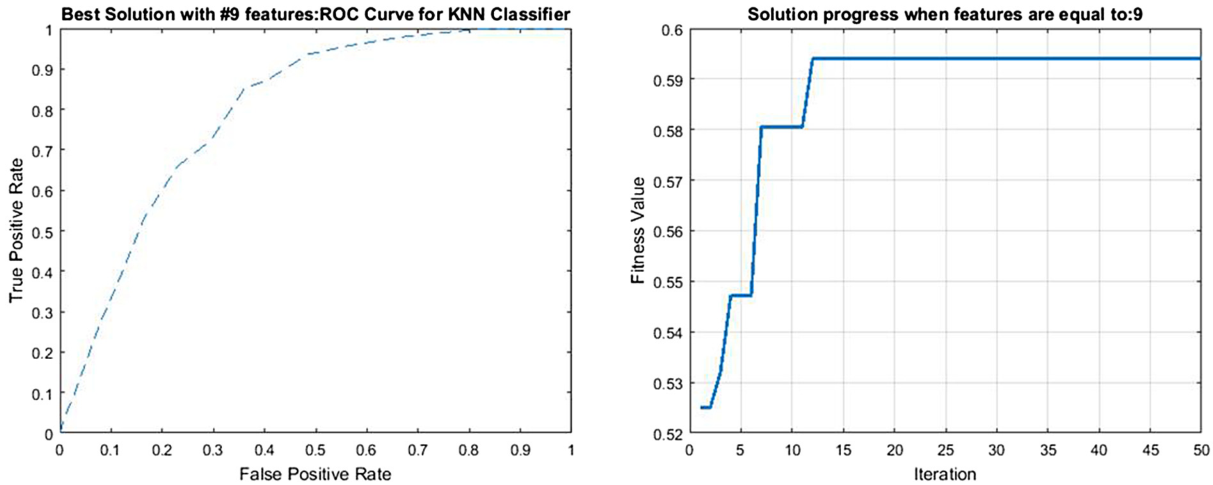

Fig. 6. ROC graph and Fitness Convergence (10-means scenario; 79.70% AUC value).

Table 16
Comparison.

Feature selection-based credit scoring approach	Feature selected (average)	AUC (average)
LDA+SVM [71]	12	0.7360
DT+SVM [71]	12	0.7120
RST+SVM [71]	12	0.7300
F-Score+SVM [71]	12	0.7450
ADE-PNN [69]	–	0.8170
ADE-LR [69]	–	0.7670
ADE-NB [69]	–	0.7970
ADE-SVM [69]	–	0.7910
DE-PNN [69]	–	0.8210
DE-LR [69]	–	0.7720
DE-NB [69]	–	0.8030
DE-SVM [69]	–	0.7890
MRMR+GA+LR [77]	11	0.8605
MRMR+GA+LR (WoE)[77]	10	0.8558
MRMR+GA+LDA [77]	12	0.7582
BA [54]	–	0.7864
CSO [54]	–	0.7860
FA [54]	–	0.7712
GS [54]	–	0.7681
GWO [54]	–	0.7843
PSO [54]	–	0.8004
WSA [54]	–	0.7916
GA+KNN [this work]	10	0.7616
GA+NB [this work]	11	0.7569

4.11. Discussion

In order to better discuss the computational results, we focus on the expert's clustering scenario that shows the best performance in terms of the AUC measure. In particular, for the dataset considered in this paper, a scenario consisting of 10 clusters was the selected one. For this scenario, we perform an additional feature selection experiment, in which a specific total number of features from all clusters should be combined and selected subject to pair-wise constraints. Different experiments were performed by varying each time the desired total number of features, whereas results related to the algorithm's classification ability, as well as the nature of the selected features, and the algorithm's running time were reported (see Table 18).

Moreover, since changing the accepted total number of features, different features may appear within the selected ones, we keep track of the frequency of appearance of each feature for interpretability purposes. The higher the frequency of appearance, the more dominant the feature is, in characterizing the classification of borrowers. Such results may lead to better qualitative understanding of the key factors driving the borrowers' ability to honor their obligations.

Table 17
Comparison (continue).

Feature selection-based credit scoring approach	Feature selected (average)	AUC (average)
ELM [53]	–	0.8930
ELM (ICS) [53]	–	0.9002
ELM (PSO) [53]	–	0.9127
ELM (BA) [53]	–	0.9167
OCHE [47]	–	0.9376
AugBoost-RP [48]	–	0.7933
AugBoost-PCA [48]	–	0.7849
AugBoost-NN [48]	–	0.7868
mg-GBDT [48]	–	0.7929
LSTM-SMOTE [50]	–	0.7828 +- 0.0386
LSTM-Improved Smote [50]	–	0.7990 +- 0.0241
GA+KNN [this work]	10	0.7616
GA+NB [this work]	11	0.7569

The progress of the AUC value as the total number of features is increased is shown in Fig. 7. It can be seen that there is a peak of the algorithm's ability for classification when nine features are combined.

Further investigation whether certain clusters do not contribute features in the final selection can be considered as an additional consistency-check of the ability of the expert to propose the right number of clusters. In the particular case reported here, the selected features originate from 8 out of 10 clusters, indicating that the initial choice of the expert proposing ten clusters was judicious. It is worth mentioning that considering the composition of each cluster, most of the features are selected from densely populated clusters.

In Fig. 8, we show the feature frequency over the whole set of experiments performed by choosing the best number of features in terms of AUC, for each scenario concerning the total proposed number of clusters. We can see that F1, F2 and F3 are selected in practically all scenarios, a fact that shows their importance in borrowers classification. It is worth noting that these features correspond to important financial quantities such as the status of existing checking account, the loan duration and the credit history, respectively. Similar comments can be made for the other frequent features.

5. Conclusions and future work

This paper introduces the idea of combining expert knowledge with evolutionary feature selection for credit risk assessment.

Table 18
Feature selection experiment with respect to the best expert's clustering scenario.

Total number of selected features	Feature subset	AUC	Computation time (HH:MM:SS)
4	F10,F16,F1,F11	0.701	01:17:22
5	F13,F14,F2,F17,F1	0.7067	01:16:19
6	F3,F15,F2,F5,F23,F1	0.7228	01:23:26
7	F3,F4,F12,F14,F21,F2,F1	0.7013	01:23:15
8	F13,F14,F21,F2,F5,F1,F7,F11	0.7386	01:30:32
9	F4,F12,F18,F2,F10,F5,F22,F9,F1	0.797	01:36:37
10	F3,F4,F15,F18,F2,F10,F16,F19,F6,F1	0.7288	01:42:52
11	F3,F18,F24,F2,F16,F17,F22,F6,F1,F7,F11	0.7697	01:47:05
12	F3,F4,F12,F13,F14,F18,F21,F2,F17,F6,F9,F1	0.749	01:55:20
13	F3,F15,F24,F2,F10,F5,F17,F6,F9,F1,F7,F8,F11	0.7089	02:04:29
14	F3,F4,F13,F15,F18,F21,F2,F5,F17,F19,F22,F6,F1,F11	0.7384	02:08:50
15	F3,F4,F13,F15,F18,F21,F10,F17,F22,F23,F6,F9,F1,F8,F11	0.7415	02:15:30

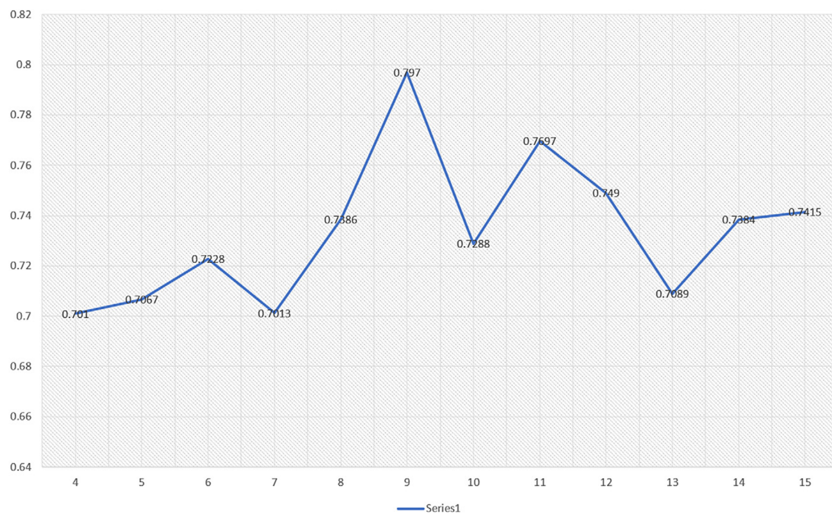

Fig. 7. AUC values with respect to clustering scenarios.

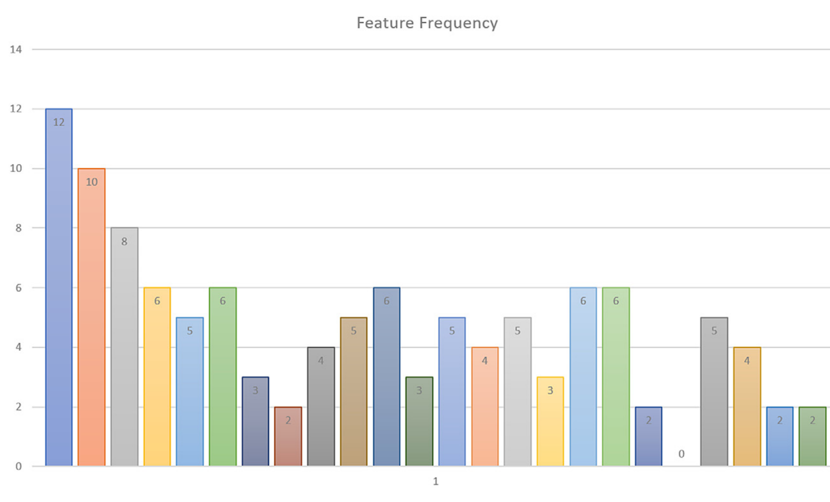

Fig. 8. Feature frequency.

The proposed solution method consists of two phases. In the first phase some feature selection-based optimization problems subject to pair-wise constraints are formulated according to an initial grouping of features (clustering) and the decision maker's pair-wise comparison judgments for each group with more than two features (AHP technique). The optimization problems depict several credit scoring scenarios that are handled via wrapper-based approaches based on the combination of the GA algorithm and the KNN or the NB supervised machine learning algorithms. The computational results achieve very good values of AUC and seem to perform well as compared to other reference algorithms.

There are several further lines of investigation. To begin with, based on the proposed scenarios (see Section 4.6), it would be interesting to create separate scenarios in which the features that already participate in pair-wise constraints could be further compared in different comparison schemas to determine how performance and significant features vary between the proposed wrapper-based approaches. Secondly, Phase I can be further updated by enclosing the option of applying Group Decision Making per cluster. For each cluster, if more than one decision makers are involved in a decision, there is a need to synthesize their opinions. In this case, there are two options available: (a) the different pair-wise comparison matrices (i.e., one for each decision maker) can be aggregated into a single pair-wise comparison matrix and then the priority vector can be calculated or (b) priority vectors can be derived from each of the different pair-wise comparison matrices and then aggregated into one priority vector. Thirdly, it would be interesting to see how the proposed wrapper approaches perform in case of imbalanced data. In this case, some modifications could take place to the proposed algorithm. For instance, a simple under-sampling technique can be used to under-sample the majority class randomly and uniformly. The flowchart for the machine learning process (see Fig. 3) could be changed as follows. After splitting the original dataset into training and test sets, the training set should be checked if it is imbalanced. In case of imbalanced data, an under-sampling technique is applied on the training set. This training set can now be further divided into training and validation sets in the context of k-fold cross validation process. Finally, Phase I can be further investigated by applying an option of finding the optimal number of clusters for the k-means algorithm (i.e., k is a decision variable of the optimization problem to be found) based on several criteria. The optimal k aims at producing a partitioning with low intra-cluster variances (compact clusters), low inter-cluster similarities and manageable number of features per cluster to be evaluated by a decision maker.

CRediT authorship contribution statement

Pantelis Z. Lappas: Conceived of the presented idea, Developed the theory and performed the computations, Carried out the experiment, Wrote the manuscript. **Athanasios N. Yannacopoulos:** Conceived of the presented idea, Developed the theory and performed the computations, Carried out the experiment, Wrote the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Background

This section describes the basic concepts related to computational intelligence tools used to construct an optimization framework for credit risk assessment.

A.1. k-means clustering

Clustering or grouping is the task of dividing a set of objects into homogeneous clusters/ groups. The k-means algorithm is one of the most popular algorithms of combinatorial cluster analysis. For details see e.g. [98]. In general, a set of m objects $\Xi = \{\xi_1, \xi_2, \dots, \xi_m\}$ is given. In the context of this paper, Ξ is a set of borrowers. Each borrower, ξ,is described by an N-dimensional feature vector $\psi_i = (\omega_{i1}, \dots, \omega_{im})^T$, where ω_{ij} denotes the value of the jth feature of the ith borrower, ξ_i. For the purpose of analysis, the set of objects may be characterized by a representation given through the matrix $\widetilde{\Psi} = [\psi_1, \dots, \psi_m]$, or rather its transpose, $\Psi = \widetilde{\Psi}^T$, whose ith row is a vector of the ith borrower's features. It will be convenient to use the following notation: given a matrix M, by $M[:, j]$ we denote the jth column, whereas by $M[i, :]$ we denote the ith row of the matrix M, respectively. Using this notation, $\Psi[i, :] = \psi_i$, are the characteristics of the ith borrower, whereas

$$\Psi[:, \ell] = [\omega_{1\ell}, \omega_{2\ell}, \dots, \omega_{m\ell}] =: \phi_\ell, \tag{A.1}$$

is the distribution of feature ℓ among the borrowers. When all features are quantitative, each customer may be visualized as a point in the space $\mathbb{R}^N$. Borrowers which are similar in features can then be visualized as clouds of nearby points in this high dimensional space. The objective of the method is to classify all borrowers into k groups (hereafter referred to as clusters) of borrowers with similar features, with respect to some suitable notion of similarity, which is usually understood as some appropriate notion of distance. The number of clusters k can either be pre-assigned or not. Similarity can be quantified in terms of an appropriate distance in the space of features. In particular, the Euclidean distance

$$d(\xi_i, \xi_j) = \|\psi_i - \psi_j\| = \sqrt{\sum_{r=1}^{N} |\omega_{ir} - \omega_{jr}|^2} \tag{A.2}$$

can be used as a measure of dissimilarity.

Let k be the (given) number of clusters, cardinality of set Ξ, $|\Xi|$, the number of objects, and N_j the number of objects to be allocated in the jth cluster ($j = 1 \cdots, k$). The membership of object i, represented by feature vector ψ_i, with cluster j to be found is denoted by the weight w_{ij} that is equal to 1 if object i ($i = 1, \dots, m$) is allocated to cluster j (0 otherwise). The cluster j is partially characterized by

$$c_j = \frac{1}{N_j} \sum_{i=1}^{|\Xi|} w_{ij} \psi_i, \quad j = 1, \dots, k, \tag{A.3}$$

which corresponds to each center (to be found). The k-means algorithm, whose objective is to assign each object to a cluster whose center it is closer to it [98], can be formulated as an optimization problem where the elements of a set of $m = |\Xi|$ objects have to be allocated to k clusters such that the sum of squared Euclidean distances between each object and the center of the cluster in which each element is assigned to is minimized [99].

$$\begin{aligned} &\min_{\{w_{ij}\}} \sum_{i=1}^{|\Xi|} \sum_{j=1}^{k} w_{ij} \|\psi_i - c_j\|^2, \quad \text{s.t.}, \\ &\sum_{j=1}^{k} w_{ij} = 1, \quad i = 1, \dots, |\Xi|, \\ &w_{ij} \in \{0, 1\}, \quad i = 1, \dots, |\Xi|, \quad j = 1, \dots, k, \end{aligned} \tag{A.4}$$

where c_j is given by (A.3).

In this paper, we take a different route, in that instead of clustering borrowers in similar groups, we use the customer population as an indication of the distribution of different features and cluster instead the features into similar groups in an attempt to find possible strong correlations between features. Such information will be very useful in case that a large number of features are available, as this set can provide an initial dimensionality reduction criterion. In particular, if two features belong to the same cluster, this is an indication that perhaps not both of them provide important independent information on the credit quality of the applicant. Therefore, knowledge of such high correlations might be useful as an initial step of dimensionality reduction. Hence, we can use the k-means clustering for grouping features of the original dataset into predefined number of clusters, according to alternative scenarios. This can be achieved by a variant of problem (A.4) where ψ_i is replaced with ϕ_ℓ, as defined in (A.1). In particular,

$$\begin{aligned} &\min_{\{w_{il}\}} \sum_{i=1}^{N}\sum_{\ell=1}^{k} w_{i\ell}\|\phi_\ell - c_\ell\|^2, \quad \text{s.t.,} \\ &\sum_{\ell=1}^{k} w_{i\ell} = 1, \quad i = 1,\dots,N, \\ &w_{i\ell} \in \{0,1\}, \quad i = 1,\dots,N \quad \ell = 1,\dots,k. \end{aligned} \tag{A.5}$$

where $c_\ell = \frac{1}{N_\ell}\sum_{i=1}^{N} w_{i\ell}\phi_\ell$, with N_ℓ the number of features to be allocated in the ℓth cluster ($\ell = 1\cdots,k$), $w_{i\ell}$ the associated weight of ϕ_i with cluster ℓ (to be found), that is equal to 1 if feature i ($i = 1\cdots,N$) is allocated to cluster ℓ (0 otherwise). The above problem consists of clustering the columns of matrix Ψ, corresponding to the distribution of features into similar groups [100]. Both optimization problems, (A.4) and (A.5), are equivalent formulations of the k-means clustering approach, in an appropriate metric space setting, which can be solved by applying the standard k-means algorithm [98].

A.2. Analytic hierarchy process

One important aspect of the present paper is the inclusion of expert opinion in the feature selection process. This is very important especially for applications related to credit scoring. As expert opinion may be subjective, it is important to use a procedure to turn expert opinion into an objective information that can be incorporated in the machine learning procedure. The AHP is ideal for this task.

The Analytic Hierarchy Process (AHP) is a decision making approach introduced by Saaty [101]. In this paper, the AHP is applied as a decision making tool to rank the relative importance of features within a given cluster as characterized by the clustering approach in Appendix A.1. This characterization allows the decision maker to submit a rating in the form of a numerical vector for a large number of alternatives (i.e., features). The main goal is to rank features within a cluster, select the most significant and set pair-wise constraints between low-score features of relatively equal importance.

Fixing a particular cluster of features ℓ, out of the k clusters obtained by the procedure described in Appendix A.1, consisting of the set of $n = N_\ell$ features, $F_\ell = \{f_{\ell_1}, f_{\ell_2}, \dots, f_{\ell_n}\} \subset F = \{f_1,\dots,f_N\}$, $n = N_\ell < N$, the decision maker should provide a weight vector $w' = [w'_1, w'_2, \dots, w'_n]^T$, where w'_i is a score value of feature f_i ($i = 1, 2\cdots,n$). These scores are assigned so that, the greater the w'_i is, the more important the ith feature is, among the other features in the same cluster, according to the preferences of the decision maker. More precisely, the decision maker is asked to state pair-wise comparisons between the features within the cluster and the AHP process is designed so as to control his/her possible inconsistencies by considering two features at a time [102]. The pair-wise comparisons are collected into a pair-wise comparison matrix,

$$A = (a_{ij})_{nn} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \tag{A.6}$$

where $a_{ij} > 0$ expresses the degree of preference of f_i with respect to f_j. The elements a_{ij}, $i > j$ are obtained by comparing the importance of features i and j in terms of the 1–9 scale [103], presented in Table A.19, whereas the lower diagonal elements a_{ji} are set as $\frac{1}{a_{ij}}$, for consistency. Clearly, the diagonal elements a_{ii} should equal 1.

Table A.19
1–9 scale for relative importance according to Saaty [103].

Degree	Description
1	Equal relative importance
2	Equal to moderate relative importance
3	Moderate relative importance
4	Moderate to strong relative importance
5	Strong relative importance
6	Strong to very strong relative importance
7	Very strong relative importance
8	Very to extremely strong relative importance
9	Extremely strong relative importance

The above scale indicates the level of relative importance from equal, moderate, strong, very strong to extreme level by 1, 3, 5, 7 and 9, respectively (i.e., odd numbers). The intermediate values between two adjacent comparisons are denoted by 2, 4, 6 and 8 (i.e., even numbers). For instance, $a_{13} = 3$ means that f_1 is moderately preferred to f_3; $a_{24} = 5$ means than f_2 is strongly preferred to f_4. Alternatively, pair-wise comparisons in this form follow the assumption that if f_1 is 3 times more important than f_3, then it can be deduced that f_3 is 1/3 as important as f_1; if f_2 is 5 times more important than f_4, then f_4 is 1/5 as important as f_1.

According to Saaty [101], it is supposed that each a_{ij} approximate the ratio between two weights: $a_{ij} = \frac{1}{a_{ji}} \simeq \frac{w'_i}{w'_j}$ for every i, j. Therefore, ideally we should be able to restructure matrix A as follows

$$\begin{aligned} A = (a_{ij})_{nn} &= \begin{bmatrix} 1 & a_{12} & \cdots & a_{1n} \\ \frac{1}{a_{12}} & 1 & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{a_{1n}} & \frac{1}{a_{2n}} & \cdots & 1 \end{bmatrix} \\ &\simeq \left(\frac{w'_{ii}}{w'_{ij}}\right)_{nn} = \begin{bmatrix} \frac{w'_1}{w_1} & \frac{w'_1}{w'_2} & \cdots & \frac{w'_1}{w'_n} \\ \frac{w'_2}{w'_1} & \frac{w'_2}{w'_2} & \cdots & \frac{w'_2}{w'_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{w'_n}{w'_1} & \frac{w'_n}{w'_2} & \cdots & \frac{w'_n}{w'_n} \end{bmatrix}. \end{aligned} \tag{A.7}$$

Eq. (A.7) motivates a connection of the weight assignment problem with the maximal eigenvalue problem for matrix A [101] and in particular the w' is the normalized eigenvector corresponding to the maximal eigenvalue, λ_{max}, of matrix A.

After collecting the decision maker's pair-wise comparisons and computing the weight vector, an evaluation of how inconsistent decision maker's judgments are, should take place [102]. A possible measure of consistency of the pair-wise comparison matrix is given by:

$$CR_A = \frac{CI_A}{RI_n}$$

where CI_A is the consistency index of A defined as

$$CI_A = \frac{\lambda_{max} - n}{n - 1},$$

and RI_n is a random index which depicts an estimation of the average CI obtained from a large enough set of randomly generated matrices A of size n. This index is presented in Table A.20. Actually, if the index CR_A is less than 0.10, the decision maker's judgments are relatively consistent.

Table A.20
Random index values according to size of matrix.

n	3	4	5	6	7	8	9	10
RI_n	0.5247	0.8816	1.1086	1.2479	1.3417	1.4057	1.4499	1.4854

For an alternative interpretation of the AHP method based on geometric mean of the elements of matrix A, as well as an alternative consistency measure see [104] and [105], respectively.

A.3. Genetic algorithm

Since a pre-selection procedure based on the clustering and the AHP approach has been applied, the credit scoring problem is solved considering a sub-set of features already evaluated by experts. The use of experts opinion in the pre-selection procedure for selecting groups of variables is extremely important especially in applications concerning credit risk. Then, the credit scoring problem is solved by using an evolutionary learning approach in order to find the most significant features for modeling and prediction purposes.

Genetic algorithms (GA) are evolutionary optimization algorithms inspired by the biological mechanisms of natural selection and reproduction [106]. GAs work throughout populations of possible solutions (i.e., population-based search meta-heuristic algorithm) known as "generations" where each solution (called individual) is represented as a finite length string (called chromosome), over some finite set of symbols (real-coded or binary). Each symbol of the finite length string is called "gene" and represents the value of the decision variable on a continuous domain (real-coded chromosome) or on a binary domain (0/1). Furthermore, the "fitness" of each individual is associated with the evaluation (or objective) function of the problem.

In the context of the feature selection problem, each chromosome represents a subset of features as a bit string (0: do not choose a given feature; 1: choose a given feature). A new generation of possible solutions is obtained from the current generation by applying suitable operators. For instance, upon choosing a pair of high performing solutions (called parents) new solutions are generated by applying (a) a crossover operator for exploring new solutions (called offspring) and exchanging information between strings and (b) a mutation operator for maintaining and introducing diversity in the population.

Fig. A.9 illustrates an example of two parent individuals (i.e., $p1$ and $p2$), their chromosome representation, as well as the operators of crossover to produce two offspring (i.e., $o1$ and $o2$) and mutation.

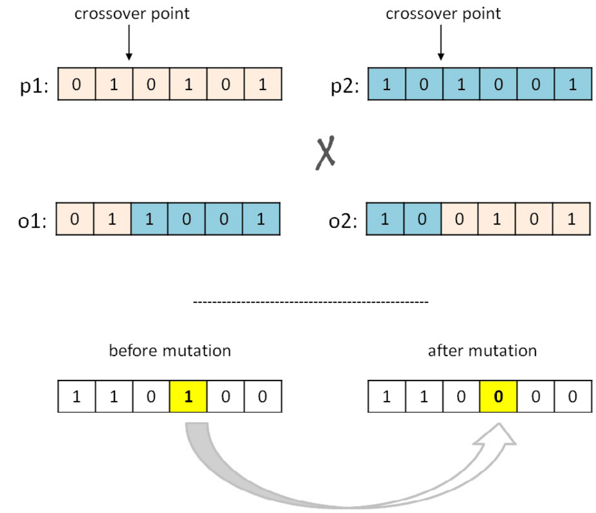

Fig. A.9. Chromosome representation, crossover and mutation operators.

In this paper, GAs are used as a wrapper method in which a chosen machine learning algorithm is trained on a previously selected subset of features and then evaluated with a chosen performance metric. Two supervised machine learning algorithms are embedded in the wrapper approach of GAs in particular, the k-Nearest Neighbors (KNN) and Naive Bayes (NB) algorithms. Ling et al. [107] and Huang & Ling [108] showed, theoretically and empirically, that if a single-number measure is to be used for evaluating the predictive ability of machine learning algorithms, the Area Under the ROC (Receiver Operating Characteristics) Curve (AUC) is a good choice. The ROC graph depicts the relationship between sensitivity (i.e., the proportion of bad borrowers correctly identified) and specificity (i.e., the proportion of good borrowers correctly identified) of a binary classification. The AUC can be any value between 0 and 1, where a value of 0 indicates a perfectly inaccurate test and a value of 1 indicates a perfectly accurate test. As a result, the higher the AUC, the better the model is in distinguishing between good and bad borrowers. The AUC performance measure is used as a fitness function in the context of GA so as to evaluate the individuals of a population.

A.4. Supervised machine learning algorithms

In general, any combination of a search strategy (i.e., selection of a subset of features) and a machine learning algorithm (i.e., evaluation of the learning algorithm on the selected subset of features) can be used as a wrapper method. KNN and NB machine learning algorithms are going to be combined with GA so as to develop and compare two wrapper methods for classifying borrowers into two groups (good and bad ones).

Assuming a sample of borrowers $\Xi = \{\psi_1, \dots, \psi_m\}$ (see Appendix A.1), hereafter called the instance domain, Eq. (A.2) is a function that returns the Euclidean distance between any two elements of Ξ. For each element ψ_i of Ξ, we have a characterization (e.g. good or bad borrower) which will be denoted by a variable c_i either numeric or binary (e.g. $c_i = 1$ or 0 for a good or bad borrower, respectively). The basic aim of any learning algorithm is to identify a suitable function that associates ψ_i with c_i. Once this function $\psi_i \mapsto c_i =: \mathcal{F}(\psi_i)$, has been fitted to the training sample Ξ, it can be used for classification or prediction of the credit behavior of newcomers, assigning to a newcomer ψ the value $c = \mathcal{F}(\psi)$. In our case, the range of $\mathcal{F}$ is a finite set denoted by $R(\mathcal{F}) = \{\tau_1, \dots, \tau_H\}$ (e.g., $\{0, 1\}$ etc.).

Let $S = (\psi_1, c_1), \dots, (\psi_m, c_m)$ be a sequence of training examples containing features (i.e., independent variables: ψ_i) and target values (i.e., dependent variables, c_i) for $i = 1, \dots, m$. For each $\psi' \in \mathbb{R}^n$, such that $\psi' \notin \Xi = \{\psi_1, \dots, \psi_m\}$, considered as a new potential borrower, we consider its distance from each existing borrower in Ξ, and re-order Ξ in ascending order with respect to this criterion. Formally, let $\pi_1(\psi'), \dots, \pi_m(\psi')$ be a reordering of $\{1, \dots, m\}$ according to the distance $d(\psi', \psi_i)$ of ψ_i, $i = 1, \dots, m$, with respect to ψ'. That is, for all $i \leq m - 1$,

$$d(\psi', \psi_{\pi_i(\psi')}) \leq d(\psi', \psi_{\pi_{i+1}(\psi')}). \tag{A.8}$$

Therefore, selecting the number of neighbors, k, the KNN rule for classifying borrowers is that for any point $\psi' \in \mathbb{R}^n$ we construct the set $\{d(\psi', \psi_{\pi_i(\psi')}) \,:\, i \leq k\}$, which is the set of distances of the k-nearest neighbors to ψ', associate the elements $\pi_i(\psi') \in \Xi$, $i \leq k$, with their corresponding target label $\tau_{\pi_i(\psi')}$, and then find the majority target label among them [94].

NB is a probabilistic classifier based on Bayes' theorem under the assumption of strong independence among the features [94]. Suppose that a subset of features to be evaluated has been selected by the GA. Note that this subset by construction satisfies the pairwise constraints set by the initial clustering phase of the method. We will denote this subset of features by $F' = \{f'_1, f'_2, \ldots, f'_{n'}\} \subset F$. According to Bayes' theorem the posterior probability $Prob(c = \tau_h \mid X)$ is calculated as follows

$$Prob(c = \tau_h \mid X) = \frac{Prob(c = \tau_h)Prob(F' \mid c = \tau_h)}{Prob(F')} \tag{A.9}$$

where probabilities $Prob(c = \tau_h)$ and $Prob(F')$ are the prior probabilities and $Prob(F' \mid c = \tau_h)$ is the likelihood. After computing all the posterior probabilities, the instance F' will be assigned to class τ_k, if and only if

$$Prob(c = \tau_k \mid F') > Prob(c = \tau_h \mid F'),\ \ h = 1, \ldots, H\ , h \neq k. \tag{A.10}$$

References

[1] L. Thomas, D. Edelman, J. Crook, Credit Scoring and Its Applications, SIAM., Philadelphia, United States of America, 2002.

[2] N. Siddiqi, Intelligent Credit Scoring: Building and Implementing Better Credit Risk Scorecards, John Willey & Sons, Inc., New Jersey, United States of America, 2017.

[3] Anderson R., The Credit Scoring Toolkit: Theory and Practice for Retail Credit Risk Management and Decision Automation, Oxford University Press, Inc., New York, Unites States of America, 2007.

[4] M. Leo, S. Sharma, K. Maddulety, Machine learning in banking risk management: A literature review, Risks 7 (29) (2019) 1–22.

[5] X. Dastile, T. Celik, M. Potsane, Statistical and machine learning models in credit scoring: A systematic literature review, Appl. Soft Comput. 91 (2020) 106263.

[6] P. Teply, M. Polena, Best classification algorithms in peer-to-peer lending, North Am. J. Econ. Finance 51 (2020) 100904.

[7] C. Luo, D. Wu, D. Wu, A deep learning approach for credit scoring using credit default swaps, Eng. Appl. Artif. Intell. 65 (2017) 465–470.

[8] C. Luo, A comprehensive decision support approach for credit scoring, Ind. Manage. Data Syst. 120 (2) (2020) 280–290.

[9] G. Teles, J. Rodrigues, S. Kozlov, Classification methods applied to credit scoring with collateral, IEEE Syst. J. 14 (3) (2020) 4557–4566.

[10] S. Fan, Y. Shen, S. Peng, Improved ML-based technique for credit card scoring in internet financial risk control, Complexity 2020 (2020) 8706285.

[11] S. Finlay, Credit Scoring, Response Modeling, and Insurance Rating: A Practical Guide for Forecasting Consumer Behavior, Palgrave Macmillan, London, United Kingdom, 2012.

[12] R. Mancisidor, M. Kampffmeyer, K. Aas, R. Jenssen, Deep generative models for reject inference in credit scoring, Knowl.-Based Syst. 196 (2020) 105758.

[13] R. Stewart, A profit-based scoring system in consumer credit: making acquisition decisions for credit cards, J. Oper. Res. Soc. 62 (9) (2011) 1719–1725.

[14] J. López, S. Maldonado, Profit-based credit scoring based on robust optimization and feature selection, Inform. Sci. 500 (2019) 190–202.

[15] K. Niu, Z. Zhang, Y. Liu, R. Li, Resampling ensemble model based on data distribution for imbalanced credit risk evaluation in P2P lending, Inform. Sci. 536 (2020) 120–134.

[16] L.-M. Junior, F.-M. Nardini, C. Renso, R. Trani, J.-A. Macedo, A novel approach to define the local region of dynamic selection techniques in imbalanced credit scoring problems, Expert Syst. Appl. 152 (2020) 113351.

[17] M. Šušteršic, D. Mramor, J. Zupan, Consumer credit scoring models with limited data, Expert Syst. Appl. 36 (3) (2009) 4736–4744.

[18] M.-J. Ariza-Garzón, J. Arroyo, A. Caparrini, M.-J. Segovia-Vargas, Explainability of a machine learning granting scoring model in peer-to-peer lending, IEEE Access 8 (2020) 64873–64890.

[19] K. Bijak, L. Thomas, Does segmentation always improve model performance in credit scoring?, Expert Syst. Appl. 39 (2012) 2433–2442.

[20] V. Djeundje, J. Crook, R. Calabrese, M. Hamid, Enhancing credit scoring with alternative data, Expert Syst. Appl. 163 (2021) 113766.

[21] P. Rashinkar, V.S. Krushnasamy, An overview of data fusion techniques, in: Conference paper presented in proceedings of the 2017 International Conference on Innovative Mechanisms for Industry Applications (ICIMIA), Bangalore, Karnataka, India, 2017, pp. 694–697.

[22] T. Meng, X. Jing, Z. Yan, W. Pedrycz, A survey on machine learning for data fusion, Inf. Fusion 57 (2020) 115–129.

[23] L. Calvet, J. de Armas, D. Masip, A. Juan, Learnheuristics: hybridizing metaheuristics with machine learning for optimization with dynamic inputs, Open Math. 15 (2017) 261–280.

[24] A. Altan, S. Karasu, S. Bekiros, Digital currency forecasting with chaotic meta-heuristic bio-inspired processing techniques, Chaos Solitons Fractals 126 (2019) 325–336.

[25] S. Karasu, A. Altan, Z. Sarac, R. Hacioglu, Predictions of solar radiation based on machine learning methods, J. Cogn. Syst. 2 (1) (2017) 16–20.

[26] A. Altan, S. Karasu, The effect of kernel values in support vector machine to forecasting performance of financial time series and cognitive decision making, J. Cogn. Syst. 4 (1) (2019) 17–21.

[27] A. Altan, S. Karasu, Recognition of COVID-19 disease from X-ray images by hybrid model consisting of 2D curvelet transform, chaotic salp swarm algorithm and deep learning technique, Chaos Solitons Fractals 140 (2020) 110071.

[28] S. Karasu, A. Altan, Recognition model for solar radiation time series based on random forest with feature selection approach. in: Conference paper presented in proceedings of the 11th International Conference on Electrical and Electronics Engineering (ELECO), Bursa, Turkey, 2019, pp. 8–11.

[29] A. Costello, A. Down, M. Mehta, Machine + man: A field experiment on the role of discretion in augmenting AI-based lending models, J. Account. Econ. 70 (2020) 101360.

[30] P. Pławiak, M. Abdar, R. Acharya, Application of new deep genetic cascade ensemble of SVM classifiers to predict the Australian credit scoring, Appl. Soft Comput. 84 (2019) 105740.

[31] F. Ferreira, S. Santos, V. Dias, An AHP-based approach to credit risk evaluation of mortgage loans, Int. J. Strateg. Prop. Manag. 18 (1) (2014) 38–55.

[32] C. Oral, Analytical hierarchy process as a tool for investment appraisal, Int. J. Econ. Finance 8 (4) (2016) 306–311.

[33] A. Khashman, Neural networks for credit risk evaluation: Investigation of different neural models and learning schemes, Expert Syst. Appl. 37 (9) (2010) 6233–6239.

[34] O. Loyola-González, Black-Box vs. White-Box: UNderstanding their advantages and weaknesses from a practical point of view, IEEE Access 7 (2019) 154096-154113.

[35] E. Rosenberg, A. Gleit, Quantitative methods in credit management: a survey, Oper. Res. 42 (4) (1994) 589–613.

[36] G. Dong, K.-K. Lai, J. Yen, Credit scorecard based on logistic regression with random coefficients, Procedia Comput. Sci. 1 (2012) 2463–2468.

[37] A. Samreen, F.-B. Zaidi, Design and development of credit scoring model for the commercial banks of Pakistan: Forecasting creditworthiness of individual borrowers, Int. J. Bus. Social Sci. 3 (17) (2012) 155–166.

[38] M. Soui, I. Gasmi, S. Smiti, K. Ghédira, Rule-based credit risk assessment model using multi-objective evolutionary algorithms, Expert Syst. Appl. 126 (2019) 144–157.

[39] Y. Peng, G. Wang, G. Kou, Y. Shi, An empirical study of classification algorithm evaluation for financial risk prediction, Appl. Soft Comput. 11 (2) (2011) 2906–2915.

[40] M. Doumpos, K. Kosmidou, G. Baourakis, C. Zopounidis, Credit risk assessment using a multicriteria hierarchical discrimination approach: A comparative analysis, European J. Oper. Res. 138 (2) (2002) 392–412.

[41] B. Baesens, D. Rösch, H. Scheule, Credit Risk Analytics: Measurement Techniques, Applications, and Examples in SAS, John Willey & Sons, Inc., New Jersey, United States of America, 2016.

[42] K. Tran, T. Duong, Q. Ho, Credit scoring model: A combination of genetic programming and deep learning, in: Proceedings of the FTC 2016 - Future Technologies Conference 2016, San Francisco, United States of America, 2016, pp. 145–149.

[43] F. Moula, C. Guotai, M.-Z. Abedin, Credit default prediction modeling: An application of support vector machine, Risk Manage. 19 (2017) 158–187.

[44] R. Goh, L. Lee, Credit scoring: A review on support vector machines and metaheuristic approaches, Adv. Oper. Res. 19 (2019) 1–30.

[45] M. Herasymovych, K. Märka, O. Lukason, Using reinforcement learning to optimize the acceptance threshold of a credit scoring model, Appl. Soft Comput. 84 (2019) 105697.

[46] B. Anderson, Using Bayesian networks to perform reject inference, Expert Syst. Appl. 137 (2019) 249–356.

[47] Y. Xia, J. Zhao, L. He, Y. Li, M. Niu, A novel tree-based dynamic heterogeneous ensemble method for credit scoring, Expert Syst. Appl. 159 (2020) 113615.

[48] W. Liu, H. Fan, M. Xia, Step-wise multi-grained augmented gradient boosting decision trees for credit scoring, Eng. Appl. Artif. Intell. 97 (2021) 104036.

[49] J. Xiao, X. Zhou, Y. Zhong, L. Xie, X. Gu, D. Liu, Cost-sensitive semi-supervised selective ensemble model for customer credit scoring, Knowl.-Based Syst. 189 (2020) 105118.

[50] F. Shen, X. Zhao, G. Kou, F. Alsaasi, A new deep learning ensemble credit risk evaluation model with an improved synthetic minority oversampling technique, Appl. Soft Comput. 98 (2021) 106852.
[51] D.-P. Fonseca, P.-F. Wanke, H.-L. Correa, A two-stage fuzzy neural approach for credit risk assessment in a Brazilian credit card company, Appl. Soft Comput. 92 (2020) 106329.
[52] V. Kuppili, D. Tripathi, D.-R. Edla, Credit score classification using spiking extreme learning machine, Comput. Intell. 36 (2020) 402–426.
[53] D. Tripathi, D.-R. Edla, V. Kuppili, A. Bablani, Evolutionary extreme learning machine with novel activation function for credit scoring, Eng. Appl. Artif. Intell. 96 (2020) 103980.
[54] R. Zhang, Z. Qiu, Optimizing hyper-parameters of neural networks with swarm intelligence: A novel framework for credit scoring, PLoS ONE 15 (6) (2020) e0234254.
[55] H. Liu, H. Motoda, Feature Selection for Knowledge Discovery and Data Mining, Springer Science and Business Media., New York, United States of America, 2012.
[56] S. Wang, J. Tang, H. Liu, Feature selection, in: C. Sammut, G. Webb (Eds.), Encyclopaedia of Machine Learning and Data Mining, Springer, New York, United States of America, 2015, pp. 503–511.
[57] R. Kohavi, G. John, Wrappers for feature subset selection, Artificial Intelligence 97 (1–2) (1997) 273–324.
[58] C. Cotta, C. Sloper, P. Moscato, Evolutionary search of thresholds for robust feature set selection: Application to the analysis of microarray data, in: G. Raidl, S. Cagnoni, J. Branke, D. Corne, R. Drechsler, Y. Jin, C. Johnson, P. Machado, E. Marchiori, F. Rothlauf, G. Smith, G. Squillero (Eds.), Applications of Evolutionary Computing, Springer, Berlin, Germany, 2004, pp. 21–30.
[59] A.-R. Hedar, J. Wang, M. Fukushima, Tabu search for attribute reduction in rough set theory, Soft Comput. 12 (9) (2008) 909–918.
[60] J. Pacheco, S. Casado, L. Núñez, A variable selection method based on tabu search for logistic regression, European J. Oper. Res. 199 (2) (2009) 506–511.
[61] S.-C. Yusta, Different metaheuristic strategies to solve the feature selection problem, Pattern Recognit. Lett. 30 (5) (2009) 525–534.
[62] W. Chen, Z. Li, J. Guo, A VNS-EDA algorithm-based feature selection for credit risk classification, Math. Probl. Eng. 2020 (2020) 4515480.
[63] J.-H. Hong, S.-B. Cho, Efficient huge-scale feature selection with speciated genetic algorithm, Pattern Recognit. Lett. 27 (2) (2006) 143–150.
[64] J. Huang, Y. Cai, X. Xi, A hybrid genetic algorithm for feature selection wrapper based on mutual information, Pattern Recognit. Lett. 28 (13) (2007) 1825–1844.
[65] K. Tan, E. Teoh, Q. Yu, K. Goh, A hybrid evolutionary algorithm for attribute selection in data mining, Expert Syst. Appl. 36 (4) (2009) 8616–8630.
[66] N. Kozodoi, S. Lessmann, K. Papakonstantinou, Y. Gatsoulis, B. Baesens, A multi-objective approach for profit-driven feature selection in credit scoring, Decis. Support Syst. 120 (2019) 106–117.
[67] M. Allam, M. Nandhini, Optimal feature selection using binary teaching learning based optimization algorithm, Journal of King Saud University - Computer and Information Sciences (2019) https://doi.org/10.1016/j.jksuci.2018.12.001.
[68] M. Taradeh, M. Mafarja, A. Heidari, H. Faris, I. Aljarah, S. Mirjalili, H. Fujita, An evolutionary gravitational search-based feature selection, Inform. Sci. 497 (2019) 219–239.
[69] G.-J. Krishna, V. Ravi, Feature subset selection using adaptive differential evolution: An application of banking, in: Conference paper presented in proceedings of the ACM India Joint International Conference on Data Science and Management of Data, Kolkata, India, 2019, pp. 157–163.
[70] H. Feng, S. Li, D. He, J. Feng, A novel feature selection approach based on multiple filters and new separable degree index for credit scoring, in: Conference paper presented in proceedings of the ACM Turing Celebration Conference, Chengdu, China, 2019, pp. 1–5.
[71] F.-L. Chen, F.-C. Li, Combination of feature selection approaches with SVM in credit scoring, Expert Syst. Appl. 37 (7) (2010) 4902–4909.
[72] S. Oreski, G. Oreski, Genetic algorithm-based heuristic for feature selection in credit risk assessment, Expert Syst. Appl. 41 (4) (2014) 2052–2064.
[73] S. Jadhav, H. He, K. Jenkins, Information gain directed genetic algorithm wrapper feature selection for credit rating, Appl. Soft Comput. 69 (2018) 11–18.
[74] D. Wang, Z. Zhang, R. Bai, Y. Mao, A hybrid system with filter approach and multiple population genetic algorithm for feature selection in credit scoring, J. Comput. Appl. Math. 329 (2018) 307–321.
[75] S.-K. Trivedi, A study on credit scoring modeling with different feature selection and machine learning approaches, Technol. Soc. 63 (2020) 101413.
[76] B. Zhu, W. Yang, H. Wang, Y. Yuan, A hybrid deep learning model for consumer credit scoring, in: Proceedings of the International Conference on Artificial Intelligence and Big Data, Chengdu, China, 2018, pp. 205–208.
[77] P.Z. Lappas, A.N. Yannacopoulos, Credit scoring: A constrained optimization framework with evolutionary feature selection, in: B. Christiansen, T. Škrinjarić (Eds.), Handbook of Research on Applied AI for International Business and Marketing Applications, IGI Global, Pennsylvania, United States of America, 2021, pp. 580–605.
[78] T. Bellotti, J. Crook, Support vector machines for credit scoring and discovery of significant features, Expert Syst. Appl. 36 (2) (2009) 3302–3308, Part2.
[79] V.-S. Ha, H.-N. Nguyen, Credit scoring with a feature selection approach based deep learning, MATEC Web Conf. 54 (2016) Article No: 05004, https://doi.org/10.1051/matecconf/20165405004.
[80] D. Boughaci, A. Alkhawaldeh, Three local searh-based methods for feature selection in credit scoring, Vietnam J. Comput. Sci. 5 (2) (2018) 107–121.
[81] D. Tripathi, D.-R. Edla, R. Cheruku, V. Kuppili, A novel hybrid credit scoring model based on ensemble feature selection and multilayer ensemble classification, Comput. Intell. 35 (2019) 371–394.
[82] J. Nalić, G. Martinović, Ž. Drago, New hybrid data mining model for credit scoring based on feature selection algorithm and ensemble classifiers, Adv. Eng. Inform. 45 (2020) 101130.
[83] S. Mahdavi, S. Rahnamayan, K. Deb, Opposition based learning: A literature review, Swarm Evol. Comput. 39 (2018) 1–23.
[84] S. Rahnamayan, H. Tizhoosh, M. Salama, A novel population initialization method for accelerating evolutionary algorithms, Comput. Math. Appl. 53 (10) (2007) 1605–1614.
[85] E.-G. Talbi, Metaheuristics: From Design To Implementation, John Willey & Sons, Inc., New Jersey, United States of America, 2009.
[86] C. Emmanouilidis, A. Hunter, J. Maclntyre, A multiobjective evolutionary setting for feature selection and a commonality-based crossover operator, in: Proceedings of the 2000 Congress on Evolutionary Computation, California, United States of America, 2000, pp. 309-316.
[87] C. Papadimitriou, K. Steiglitz, Combinatorial Optimization: Algorithms and Complexity, Dover Publications, Inc., Mineola, New York, United States of America, 1982.
[88] M. Mitchell, Complexity: A Guided Tour, Oxford University Press, Oxford, United Kingdom, 2009.
[89] M. Pakhira, A linear time-complexity k-means algorithm using cluster shifting, in: Conference paper presented in proceedings of the 6th International Conference on Computational Intelligence and Communication Networks, Bhopal, India, 2014, pp. 1049–1053.
[90] Z. Hira, D. Gillies, A review of feature selection and feature extraction methods applied on microarray data, Adv. Bioinform. 2015 (2015) 198363.
[91] A. Tung, Time complexity and consistency issues in using the AHP for making group decisions, J. Multi-crit. Decis. Anal. 7 (1998) 144–154.
[92] N.-J. Mamat, J. Daniel, Statistical analyses on time complexity and rank consistency between singular value decomposition and duality approach in AHP: A case study of faculty member selection, Math. Comput. Modelling 46 (2007) 1099–1106.
[93] S. Marsland, Machine Learning: An Algorithmic Perspective, Taylor & Francis Group, LLC, Florida, United States of America, 2015.
[94] M. Kubat, An Introduction To Machine Learning, Springer International Publishing., Cham, Switzerland, 2015.
[95] S. Siraj, L. Mikhailov, J. Keane, Priest: an interactive decision support tool to estimate priorities from pair-wise comparison judgements, Int. Trans. Oper. Res. 22 (2) (2015) 217–235.
[96] T. Hastie, R. Tibshirani, J. Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Springer., New York, United Stated of America, 2008.
[97] L. Breiman, J. Friedman, R. Olshen, C. Stone, Classification and Regression Trees, Chapman & Hall/CRC, Taylor & Francis Group., United States of America, 1984.
[98] S. Wierzchoń, M. Kłopotek, Modern Algorithms of Cluster Analysis, Springer International Publishing, Cham, Switzerland, 2018.
[99] Y. Marinakis, M. Marinaki, M. Doumpos, N. Matsatsinis, C. Zopounidis, A hybrid ACO-GRASP algorithm for clustering analysis, Ann. Oper. Res. 188 (2011) 343–358.
[100] D.-P. Ismi, S. Panchoo, M. Murinto, K-means clustering based filter feature selection on high dimensional data, Int. J. Adv. Intell. Inform. 2 (1) (2016) 38–45.
[101] T. Saaty, The Analytic Hierarchy Process: Planning, Priority Setting, Resource Allocation, McGraw-Hill, New York, United States of America, 1980.
[102] M. Brunelli, Introduction To the Analytic Hierarchy Process, Springer, New York, United States of America, 2015.

[103] T. Saaty, A scaling method for priorities in hierarchical structures, J. Math. Psych. 15 (3) (1977) 234–281.
[104] G. Crawford, C. Williams, A note on the analysis of subjective judgment matrices, J. Math. Psycol. 29 (4) (1985) 387–405.
[105] J. Aguarón, J.-M. Moreno-Jiménez, The geometric consistency index: Approximated thresholds, European J. Oper. Res. 147 (1) (2003) 137–145.
[106] M. Mitchell, An Introduction To Genetic Algorithms, MIT Press, Massachusetts, United States of America, 1998.
[107] C. Ling, J. Juang, H. Zhang, in: Y. Xiang, B. Chaib-draa (Eds.), AUC: A Better Measure than Accuracy in Comparing Learning Algorithms, Springer, Berlin, Germany, 2003, pp. 329–341.
[108] J. Huang, C. Ling, Using AUC and accuracy in evaluating learning algorithms, IEEE Trans. Knowl. Data Eng. 17 (2005) 299–310.